JOB MARKET PAPER

Of Fathers and Farms

How Intergenerational Persistence Shapes Structural Transformation

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Abstract. The transmission of skills and preferences from parents to children leads to intergenerational persistence in sector choice. How do these inherited connections alter the process of structural transformation? Using linked father–son records in census microdata, this paper establishes that both inherited skills and inherited preferences are important sources of persistence in the United States between 1880 and 1940. Embedding these findings into a macroeconomic model of structural change, the paper shows that intergenerational persistence slowed the process of U.S. industrialization by 14.3 percent. Inherited skills account for nearly all of this slowdown because their immediate and dynamic effects reinforce each other, whereas those of inherited preferences largely offset. Accounting for persistence also implies substantially greater agricultural technology growth relative to nonagriculture than a model without persistence would infer from the same historical transition.

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Introduction

Most people today know a doctor, a lawyer, or an academic who followed in their parents' footsteps. Occupational transmission is not limited to high-skilled professions; indeed, it is especially pronounced among farmers. This micro-level fact raises a macro-level question: how does intergenerational persistence in sector choice reshape the dynamics of structural transformation?

Research on structural change has typically focused on technological and demand forces that reallocate labor across sectors (Herrendorf, Rogerson and Valentinyi, 2014). Recent work has highlighted new generations of workers entering the labor force as a critical source of this reallocation (Hobijn, Schoellman and Vindas, 2018; Porzio, Rossi and Santangelo, 2022; Adão, Beraja and Pandalai-Nayar, 2024). Intuitively, a new worker may be more flexible to changing economic conditions than an incumbent who has formed strong ties through their experience in a given sector. Although new workers have no employment history of their own, they nevertheless enter the labor force connected to their parents' sectors through the skills and preferences they inherit. This paper studies how these inherited connections alter the reallocation of workers out of agriculture.

As a canonical episode of structural transformation, the United States' industrialization between 1880 and 1940 provides an ideal setting to quantify the paper's insights. Using linked father–son records from the full-count historical U.S. census microdata (Helgertz et al., 2024), I document large differences in sector choice between workers whose fathers worked in agriculture (i.e., "ag-born") and those whose fathers worked outside agriculture (i.e., "nonag-born"). These differences remain substantial after accounting for childhood school attendance and local labor markets: in 1880, ag-born workers were over 3.4 times as likely to work in agriculture as nonag-born workers. Moreover, ag-born and nonag-born workers have systematically different earnings within the same sector, suggesting that workers' productivity may also depend on their parents' sector.

Guided by these empirical patterns, I develop a two-sector macro model of structural transformation with intergenerational persistence. New workers choose between agriculture and nonagriculture based on the earnings and nonpecuniary benefits available in each sector. There are two sources of intergenerational persistence: higher productivity in their parents' sector (i.e., *inherited skills*) and nonpecuniary attachments to their parents' sector (i.e., *inherited preferences*). Inherited skills increase the earnings workers can obtain in their parents' sector, while inherited preferences make agricultural employment more appealing to ag-born workers even when they could earn more elsewhere. In the model,

structural change originates from exogenous changes in either sectoral production technologies or the nonpecuniary sectoral benefits common to all workers. I model consumer demand as in Comin, Lashkari and Mestieri (2021), so technological progress reallocates labor through changes in income and relative prices.

The model shows that intergenerational persistence alters how workers respond to economic changes both within and across generations. Both inherited skills and inherited preferences reduce the aggregate responsiveness of new workers to wage changes, dampening their reallocation out of agriculture. Through these inherited connections, exogenous factors pulling workers out of agriculture today also have dynamic effects on the next generation of workers. Inherited preferences reinforce the initial movement out of agriculture, whereas inherited skills can generate large enough wage adjustments to reverse it. Overall, how much intergenerational persistence accelerates or slows industrialization depends on the strength of both inherited skills and preferences.

Quantifying these effects for U.S. industrialization requires identifying the two sources of intergenerational persistence and inferring the economic forces driving the historical transition. Earnings differences between ag-born and nonag-born workers in the same sector reflect both inherited skills and gains from self-selection (Lagakos and Waugh, 2013). The model allows me to account for these selection effects using workers' sector choices and skill dispersion inferred from earnings variation within sector-by-parental-sector groups. Assuming workers face the same sectoral wages regardless of parental background, the remaining earnings differences identify inherited skills, while residual persistence in sector choice identifies inherited preferences. Finally, I infer the historical evolution of sectoral technologies and common nonpecuniary benefits by matching observed agricultural employment shares, relative prices, and real GDP per capita growth.

Using the historical microdata, I find that both inherited skills and inherited preferences are quantitatively large and important sources of intergenerational persistence. My estimates imply that ag-born workers are 60 percent more productive in agriculture while nonag-born workers are 80 percent more productive in nonagriculture. Together, inherited skills in both sectors account for nearly three-quarters of the observed intergenerational persistence in sector choice. I attribute the remaining one-quarter of intergenerational persistence to inherited preferences. I infer that ag-born workers choose to remain in agriculture even when that implies an earnings discount equal to 33 percent of their potential nonagricultural earnings.

I conduct counterfactual simulations to quantify how U.S. industrialization would have

unfolded without inherited skills or inherited preferences. Each counterfactual economy is calibrated to match the same aggregate moments in 1880, and the simulated economies are then exposed to the same inferred growth in sectoral technologies and common nonpecuniary benefits through 1940. Removing intergenerational persistence altogether increases industrialization by 5.5 percentage points (i.e., 14.3 percent of the observed reallocation of workers into nonagriculture).[1] Inherited skills account for nearly all of this effect. Without inherited preferences, industrialization increases by only 0.8 percentage points while removing inherited skills alone increases it by 5.9 percentage points.

To understand why inherited skills account for most of the slowdown in industrialization, I distinguish between the *immediate* response of new workers to changing economic conditions and the *dynamic* effects transmitted through subsequent generations. Both inherited mechanisms reduce new workers' aggregate responsiveness to wages; in the 1880 U.S. economy, intergenerational persistence lowers this elasticity by 48 percent. After accounting for general equilibrium feedback, inherited preferences impose a slightly greater immediate constraint on reallocation than inherited skills. However, across generations, the two mechanisms have sharply different consequences. With inherited preferences, changes in parental composition lead to further reallocation out of agriculture, mostly offsetting the immediate slowdown. With inherited skills, the resulting shifts in sectoral labor productivity induce wage adjustments that instead amplify the slowdown. Thus, the immediate and dynamic effects of inherited skills reinforce each other, while those of inherited preferences largely offset each other.[2]

Finally, I show that accounting for intergenerational persistence changes what the model infers about the forces behind U.S. industrialization. As industrialization reduces the share of ag-born workers, inherited skills shift the supply of productive labor toward nonagriculture even without changes in sectoral technologies. Without accounting for this effect, matching the observed transition requires substantially more growth in nonagricultural relative to agricultural technology: 82 rather than 48 log points, an increase of 70 percent. Thus, parent sector dependence changes not only how workers respond to historical forces, but also the technological changes the model infers to explain the observed transition.

[1] For scale, Brandt, Hsieh and Zhu (2008) find that improvements in rural labor mobility contributed 8 (out of 38) percentage points to China's industrialization between 1978 and 2004.

[2] Under alternative calibrations with modest intergenerational persistence driven primarily by inherited preferences, the dynamic effects can more than offset the immediate dampening of reallocation. In such cases, inherited preferences modestly *accelerate* structural transformation.

Literature This paper bridges two vast yet distinct literatures: research on structural transformation and research on intergenerational occupational choice. I contribute to a growing strand emphasizing the critical role of new workers for labor reallocation as well as a nascent strand highlighting the macroeconomic consequences of micro-level occupational persistence. In addition, my work yields an insight broadly relevant for research in macroeconomics and international trade that uses quantitative models of labor allocation built on extreme-value discrete choice.

Relative to the recent work showcasing the importance of new generations of workers for structural change (Hobijn, Schoellman and Vindas, 2018; Porzio, Rossi and Santangelo, 2022; Adão, Beraja and Pandalai-Nayar, 2024), I show that individual workers' inherited connections to their parents' sectors meaningfully alter the aggregate process of reallocation.[3] Quantitative macroeconomic studies of structural change have almost exclusively abstracted from this mechanism. A prominent exception is Hayashi and Prescott (2008), who attribute Japan's depressed prewar output to an institutional barrier that kept agricultural employment roughly constant by forcing the designated heir to remain on the family farm. Relative to their aggregate restriction, my framework models persistence through worker-level sector choices and allows it to arise from either inherited skills or preferences, which I discipline using microdata. Crucially, I show that these two mechanisms have very different implications for structural transformation.

There is a very large literature measuring intergenerational persistence in occupation choice and evaluating specific determinants (Abramitzky, Boustan and Matiashvili, 2025). I contribute to a growing set of studies that focus on the macroeconomic consequences of this micro-level persistence.[4] Cassan, Keniston and Kleineberg (2021), Lo Bello and Morchio (2022), and Almgren, Kramer and Sigurdsson (2025) develop quantitative models to distinguish the sources of intergenerational occupational persistence and evaluate their implications for the allocation of talent and aggregate welfare. In contrast, I ask how persistence alters the long-run reallocation of workers in an economy experiencing major changes in economic fundamentals.

Finally, my analytical result speaks to a broader literature in quantitative macroeconomics and international trade that studies worker reallocation with extreme-value discrete choice models (Artuç, Chaudhuri and McLaren, 2010; Caliendo, Dvorkin and Parro, 2019). I show that even when different groups of workers have identical labor supply

[3] Kim and Topel (1995) and Lee and Wolpin (2006) also remark on new workers' role in labor reallocation.

[4] Several empirical papers focus on intergenerational persistence in agriculture and remark on the potential importance for structural change without explicitly studying it (Emran and Shilpi, 2011; Fernando, 2022; Haws, Just and Price, 2025; Liu, 2026).

elasticities, the aggregate labor supply elasticity can be substantially lower if these groups differ in their sector choices. In my setting, intergenerational persistence generates precisely these differences between ag-born and nonag-born workers. Moreover, because the composition of these groups changes across generations, so does the aggregate responsiveness of workers to wages.[5]

Roadmap Section 1 presents the motivating evidence on intergenerational persistence in the U.S. between 1880 and 1940. Section 2 develops a partial equilibrium Roy-type model of sector choice with intergenerational persistence and uses it to measure inherited productivity and immobility wedges. Section 3 embeds the partial equilibrium model into a general equilibrium environment and studies how intergenerational persistence shapes the dynamic evolution of worker allocations in response to changing economic fundamentals. Section 4 calibrates the model to the U.S. economy and quantifies the macroeconomic consequences of intergenerational persistence for its structural transformation in the first half of the 20th century. Section 5 concludes.

1. Motivating Facts

In this section, I establish several facts about the sectoral allocation of new workers in the early 20th-century U.S. economy. I show that the sons of farmers (i.e., "ag-born workers") are substantially more likely to work in agriculture than the sons of nonfarmers (i.e., "nonag-born workers"). This gap remains large after conditioning on childhood schooling and both childhood and adulthood location, and it is much larger in agriculture than in other sectors. I then show that ag-born and nonag-born workers have systematically different earnings within the same sector. Together, these facts motivate the next section's Roy-type model where workers may be more productive or otherwise more attached to their parents' sectors.

To document these facts, I use full-count U.S. decennial census microdata for 1880–1940 available from Integrated Public Use Microdata Series (Ruggles et al., 2024). The data contains demographic and limited employment information for every individual in the United States at the time of each census. I restrict the data to U.S.-born men aged 20–29 who are working and whose industry is observed in the census. Specifically, I

[5]Relatedly, Böhm, Etheridge and Irastorza-Fadrique (2025) examine how the mobility patterns of incumbent workers shape occupational labor supply elasticities, whereas my result emphasizes how differences in sector choice among groups of new workers shape their aggregate labor supply responsiveness.

classify workers as agricultural if their industry is classified by IPUMS as agriculture[6] and nonagricultural otherwise. In order to measure intergenerational persistence, I must observe each worker's parental sector. To this end, I use the IPUMS Multigenerational Longitudinal Panel (Helgertz et al., 2024) to link new entrants to their fathers in earlier censuses.[7] Historical census research has developed several alternative procedures for linking individuals across censuses.[8]

Fact 1: High intergenerational persistence during U.S. industrialization. In Figure 1, I plot the selected samples' agricultural employment share in each census year. Two features stand out. First, the aggregate share of new workers employed in agriculture falls sharply over the period, from 61 percent in 1880 to 23 percent in 1940. Second, the sons of farmers and the sons of nonfarmers allocate themselves very differently across sectors. In 1880, over 80 percent of ag-born workers are employed in agriculture compared to less than 20 percent of nonag-born workers.

Fact 2: Persistence is robust and especially pronounced for agriculture. Consider the following linear probability model estimated among individuals i in census year t:

$$\mathbb{1}\{\text{Sector = Agriculture}\}_{i,t} \quad = \quad \beta_0 \quad + \quad \beta_1 \cdot \mathbb{1}\{\text{Father's sector = Agriculture}\}_{i,t} \quad + \quad \varepsilon_{i,t}$$

Without any additional covariates, the estimated coefficient β_1 exactly equals the difference in agricultural employment rates between ag-born and nonag-born workers as depicted in Figure 1. In 1880, that unconditional gap is 66.7 percentage points. The gap attenuates slightly over time but remains large throughout the period, falling to 57.5 percentage points by 1940. To ensure that this gap is not entirely driven by differences in access to schooling or variation across local labor markets, I estimate additional specifications that include controls for childhood schooling and both childhood and adulthood commuting-zone fixed effects. The estimated β_1 coefficients from these specifications are plotted in Figure 2.

Schooling is an essential way that workers develop human capital during their youth,

[6] IND1950 codes 100–199.

[7] In particular, I use the MLP to link individuals to their own childhood records in earlier censuses, and then use the intra-census household links to identify their fathers and their fathers' industries.

[8] Appendix A.1.2 shows that my results are robust to using links from the Census Tree (Buckles et al., 2025) and the Census Linking Project (Abramitzky et al., 2022). The former combines genealogy-based and genealogy-trained machine-learning links with links from the MLP and Census Linking Project. The latter uses rule-based matching based on individuals' names, ages, and birthplaces.

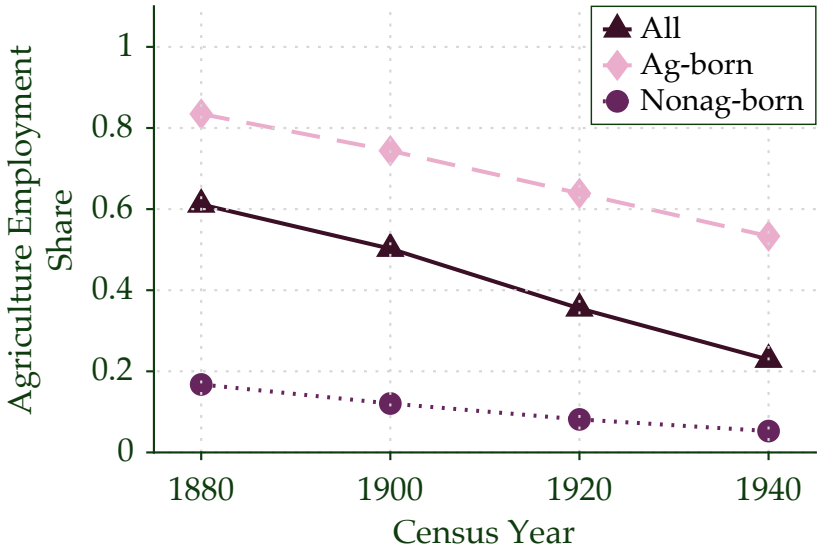


Figure 1: Industrialization of New U.S. Workers, 1880–1940

Notes: The figure plots agricultural employment shares among linked U.S.-born employed men ages 20–29 with observed own and paternal industry. Ag-born workers have fathers employed in agriculture in the earlier childhood census, while nonag-born workers have fathers employed outside agriculture. Agriculture is defined by IND1950 codes 100–199, and the "All" series pools ag-born and nonag-born workers in the linked sample.

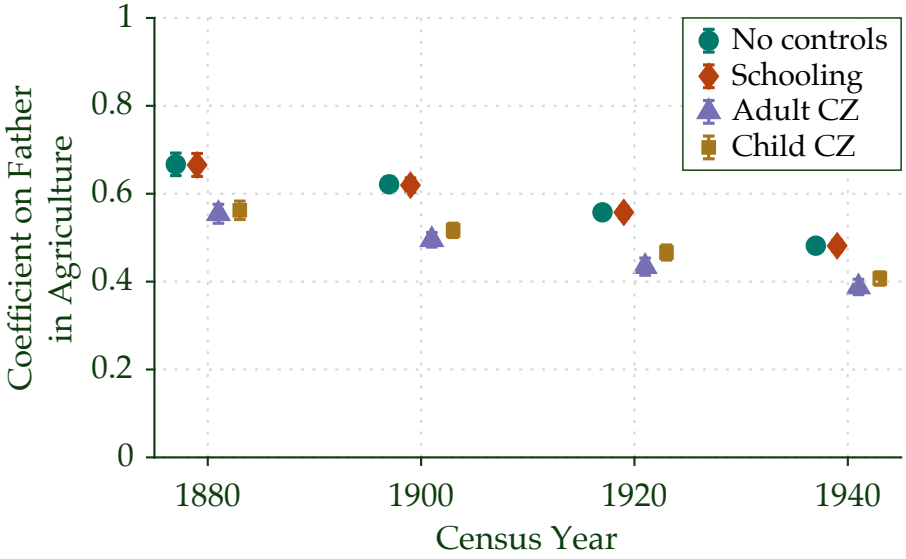


Figure 2: Intergenerational Persistence in Agricultural Employment, 1880–1940

Notes: Each marker reports the coefficient on an indicator for whether a worker's father works in agriculture from a separate linear probability model whose dependent variable indicates whether the worker works in agriculture. The four specifications include, respectively, no additional controls, childhood schooling fixed effects, adulthood commuting-zone fixed effects, and childhood commuting-zone fixed effects. The sample consists of linked U.S.-born employed men ages 20–29 with observed own and paternal industry, and agriculture is defined by IND1950 codes 100–199. Whiskers denote 95 percent confidence intervals based on standard errors clustered by adulthood commuting zone.

which may be a critical factor in determining their future occupational choices (Caselli and Coleman II, 2001; Porzio, Rossi and Santangelo, 2022). In order to determine whether the observed intergenerational persistence results from varied access to schooling for the sons of farmers and the sons of nonfarmers, I estimate a specification that includes an indicator for whether the worker was attending school when observed during their childhood. I find that the schooling-conditional gap in agricultural employment between ag-born and nonag-born workers is effectively unchanged from the unconditional gap. Guided by this finding, I will abstract from schooling in the subsequent analysis.[9]

Another natural concern is that this persistence reflects spatial differences in sectoral opportunity. If ag-born workers grow up in agrarian hinterlands and nonag-born workers in industrialized cities, their occupational divergence may simply reflect the local labor markets available to them. Recent work by Eckert, Juneau and Peters (2023) demonstrates that U.S. industrialization was highly localized, with 60–90 percent of reallocation occurring within commuting zones. As such, I test whether the large intergenerational persistence gap remains after accounting for the commuting zones where workers grew up and where they live as adults. I find that differences across observed childhood and adult local labor markets account for relatively little of the intergenerational association.

Finally, I test whether the observed intergenerational persistence is specific to agriculture or instead reflects a more general pattern of persistence across sectors. I compute the agricultural employment gap between ag-born workers and nonag-born workers whose fathers were employed in manufacturing or services, respectively. I find that these gaps are effectively the same as reported in Figure 3, suggesting that the agricultural employment gap is not driven by the particular mix of parental industries represented among nonag-born workers.

I also test whether intergenerational persistence is as strong within the nonagricultural sector. I compute the gap in manufacturing employment between workers whose fathers were employed in manufacturing or services. I find that the estimated gap within nonagriculture is substantially smaller than the gap between ag-born and nonag-born workers, suggesting that intergenerational persistence is especially pronounced for agriculture.

Fact 3: Workers' earnings differ systematically by parental sector. Workers are not only over-represented in their parents' sectors, but they also have systematically different

[9]This exercise establishes robustness to recorded childhood school attendance; it does not rule out broader schooling mechanisms. The indicator is observed at one point in childhood and does not capture cumulative attendance, educational attainment, or school quality.

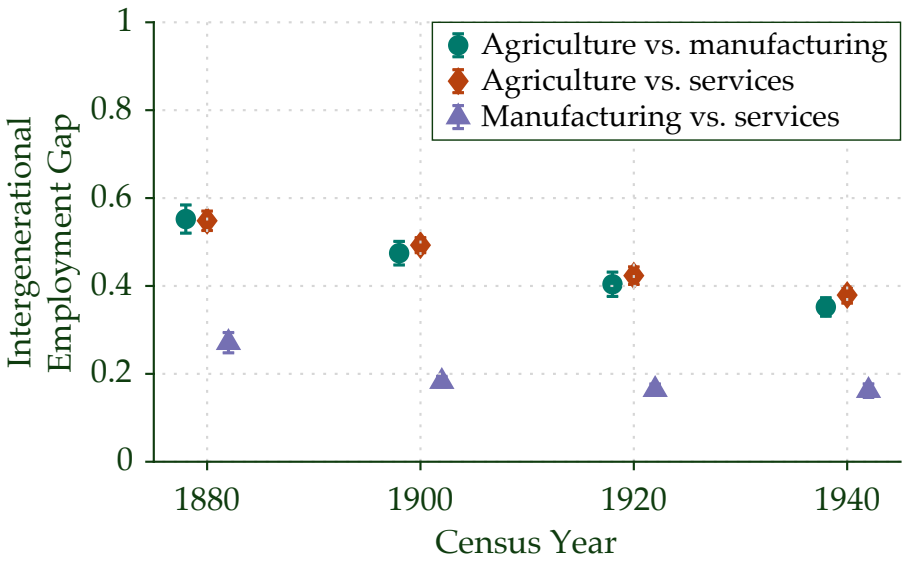


Figure 3: Intergenerational Persistence Across Sectors, 1880–1940

Notes: Each marker reports a coefficient from a separate linear probability model estimated in the indicated census year. The agriculture-versus-manufacturing and agriculture-versus-services series report the coefficient on an indicator for whether the father works in agriculture, with the comparison group restricted as indicated. The manufacturing-versus-services series restricts the sample to sons of nonagricultural fathers and reports the coefficient on an indicator for whether the father works in manufacturing. Manufacturing is defined by IND1950 codes 300–499, while services include all other nonagricultural industries. All specifications include childhood-schooling-by-commuting-zone fixed effects. Whiskers denote 95 percent confidence intervals based on standard errors clustered by adulthood commuting zone, and the sample otherwise follows Figure 2.

earnings within the same sector. The first census to collect earnings information was 1940, so I focus on that year to study the relationship between parental sector and workers' earnings. To do so, I further restrict the sample to workers with positive wage and salary income, at least 40 weeks worked, and at least 35 usual weekly hours.[10]

Table 1 reports regressions of log hourly earnings on an indicator for whether a worker's father worked in agriculture, estimated separately among agricultural and nonagricultural workers. The first two columns report unconditional gaps, while the final two columns add fixed effects for schooling, childhood commuting zone, and adulthood commuting zone, respectively.

Workers' earnings vary tremendously within their sector of employment, and their parents' sector explains a substantial portion of that. Without controls, ag-born workers earn approximately 49 percent less than nonag-born workers in agriculture and 28 percent less in nonagriculture. These gaps fall to approximately 35 and 16 percent, respectively, after conditioning on childhood schooling and both childhood and adulthood location. At first

[10]These restrictions are standard in the literature though there are some nontrivial implications for the analysis that I discuss later.

Table 1: Systematic Earnings Differences by Parent Sector, 1940

	No controls		Full controls	
	Agricultural workers	Nonagricultural workers	Agricultural workers	Nonagricultural workers
Father in agriculture	-0.677***	-0.322***	-0.436***	-0.173***
	(0.0364)	(0.0099)	(0.0150)	(0.0060)
Childhood-schooling fixed effects	No	No	Yes	Yes
Childhood-CZ fixed effects	No	No	Yes	Yes
Adulthood-CZ fixed effects	No	No	Yes	Yes
Observations	237,659	1,839,594	237,657	1,839,594
R^2	0.078	0.048	0.235	0.169
Residual variance	0.785	0.329	0.655	0.288

Notes: The dependent variable is log hourly wage and salary earnings, and the reported coefficient corresponds to an indicator for whether the worker's father worked in agriculture. Each regression is estimated separately for workers employed in agriculture and nonagriculture, as indicated by the column heading. Full controls comprise separate fixed effects for childhood school attendance, childhood commuting zone, and adulthood commuting zone. The sample consists of linked U.S.-born men ages 20–29 with valid own and paternal industries, positive wage and salary income, at least 40 weeks worked, and at least 35 usual weekly hours. Standard errors clustered by adulthood commuting zone are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

glance, it may seem surprising that ag-born workers earn less than nonag-born workers in agriculture, but this is consistent with the fact that few nonag-born workers enter agriculture and those who do are highly positively selected. In order to infer whether workers inherit productive advantages in their parents' sector, I need to separate the effects of inherited productivity from selection into sectors. The model introduced in the next section provides a clear framework to do so.

Discussion These findings establish intergenerational persistence in sectoral employment as a quantitatively important feature of the U.S. economy during this period of structural transformation. Schooling and local labor-market conditions explain little of the relationship between fathers' and sons' sectors, while parental sector also predicts earnings within destination sectors. The historical evidence therefore points to unobserved heterogeneity transmitted across generations but does not by itself reveal its source. The sons of farmers may remain in agriculture because they inherit a comparative advantage in that sector or because they inherit other attachments that reduce their incentives to leave it. The next section formalizes these forces as inherited productivity and inherited immobility wedges in a two-sector Roy model with heterogeneously skilled workers.

Combining the model's structure with workers' sector choices and earnings allows me to distinguish them and, in subsequent sections, quantify how each shaped the U.S. economy's transition out of agriculture.

2. Modeling Sector Choice with Intergenerational Persistence

I build on a Roy (1951) model of comparative advantage and self-selection, applied to sorting between agriculture and nonagriculture as in Lagakos and Waugh (2013). Workers have heterogeneous skills for two sectors of production: agriculture and nonagriculture. They choose their sector to maximize indirect utility, which depends on their earnings and sector-specific utility shifters.[11] I extend the standard framework to allow workers to inherit productive advantages (i.e., *inherited productivity*) or other attachments (i.e., *inherited immobility*) in their parents' sectors.[12] I model inherited productivity as a wedge that rescales workers' skills in their parents' sectors, while inherited immobility is a wedge that reduces ag-born workers' indirect utility from working in nonagriculture.[13]

By assuming that workers' skills are Fréchet distributed, I can derive closed-form expressions for the fractions of ag-born and nonag-born workers who choose each sector, as well as the average skill they supply to each sector. I leverage these expressions to identify the model's four labor supply parameters—the Fréchet shape parameter, the inherited productivity wedges, and the inherited immobility wedge—from four moments concerning workers' income and sectoral allocations. Taking the model to data, I use the linked father–son census records described previously to infer these parameters. In the next section, I also exploit the tractability of the Fréchet distribution to derive clear analytical insights into how these wedges influence worker reallocation.

[11] The utility shifters capture the idea that workers may have incentives for pursuing certain sectors beyond just the wage they can earn. For example, Caselli and Coleman II (2001) (and various papers thereafter including Alonso-Carrera and Raurich (2018); Porzio, Rossi and Santangelo (2022)) consider costs of pursuing nonagricultural work that workers incur but firms do not.

[12] Related evidence attributes occupational persistence to the transmission of occupation-specific human capital, access to employers and professions, information, and exposure (Laband and Lentz, 1983; Corak and Piraino, 2011; Aina and Nicoletti, 2018; Bell et al., 2019; Haeck and Laliberté, 2025).

[13] This approach is similar to Hsieh et al. (2019) who utilize a Roy model with wedges to study how discrimination, barriers to forming human capital, and social norms influence the (mis)allocation of talent across occupation for the late 20th century United States.

2.1. Environment

There is a unit mass of households indexed by i and two sectors of production indexed by $j \in \{A, N\}$, where A and N denote agriculture and nonagriculture, respectively. Households are dynastic and workers live for a single period: the worker inhabiting household i in period $t-1$ is the parent of the worker inhabiting household i in period t. As such, workers are indexed by their parent's sector of employment in the previous period, $k \in \{A, N\}$. Accordingly, refer to workers whose parents worked in agriculture as ag-born and those whose parents worked in nonagriculture as nonag-born.

Heterogeneous Sectoral Skills Workers draw i.i.d. skills, $\epsilon_{i,j,t}$, for each sector $j \in \{A, N\}$ from the stationary Fréchet distribution:

$$\epsilon_{i,j,t} \sim F(\epsilon) \equiv \exp\left[-\epsilon^{-\gamma}\right]$$

The shape parameter $\gamma > 1$ governs the inverse of skill dispersion: as $\gamma \to 1$ skills become more dispersed and as $\gamma \to \infty$ the skill distribution collapses to a single point.

Inherited Productivity Wedges Each worker inherits a productivity wedge, $\kappa_j \geq 1$, which rescales their skill for their parent's sector. Then, the talent provided by each worker to a given sector is given by:

$$h_{i,j,t} = \epsilon_{i,j,t} \cdot \begin{cases} \kappa_j & \text{if} \quad s_{i,t-1} = j \\ 1 & \text{otherwise} \end{cases}$$

Sectoral Amenities and the Inherited Immobility Wedge Working in sector j provides a common sectoral payoff shifter, $B_{j,t} > 0$, which scales workers' indirect utility without affecting their productive talent or labor income. Each ag-born worker also inherits an immobility wedge, $\tau \in [0, 1)$, which reduces their indirect utility from working in nonagriculture relative to nonag-born workers of equal skill. Let $V_{i,j,t}$ denote the indirect utility of worker i from working in sector j in period t. Then, the indirect utility of working in sector j is given by:

$$V_{i,j,t} = B_{j,t} \cdot w_{j,t} \cdot h_{i,j,t} \cdot \begin{cases} (1-\tau) & \text{if} \quad s_{i,t-1} = A \text{ and } j = N \\ 1 & \text{otherwise} \end{cases}$$

where $y_{i,j,t} \equiv w_{j,t} \cdot h_{i,j,t}$ is the labor income of worker i in sector j and $w_{j,t}$ is the wage per efficiency unit in sector j that workers take as given.

Worker's Problem Workers choose their sector to maximize indirect utility:

$$s_{i,t} \quad = \quad \underset{j \in \{A,N\}}{\arg\max} \; V_{i,j,t}$$

2.2. Optimization

Workers choose to work in agriculture if and only if:

$$\frac{\epsilon_{i,A,t}}{\epsilon_{i,N,t}} \quad \cdot \quad \begin{cases} \kappa_A \\ \frac{1}{\kappa_N} \end{cases} \quad \geq \quad \frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \quad \cdot \quad \begin{cases} (1-\tau) & \text{if} \quad s_{i,t-1} = A \\ 1 & \text{if} \quad s_{i,t-1} = N \end{cases}$$

The left-hand-side of the inequality represents workers' comparative advantage, while the right-hand-side represents the indirect utility threshold around which they sort. All else equal, the inherited agricultural productivity wedge κ_A and immobility wedge τ increase ag-born workers' incentives to choose agriculture. Likewise, the inherited nonagricultural productivity wedge κ_N raises nonag-born workers' incentives to choose nonagriculture.

Sector Allocations of Ag-born versus Nonag-born Workers The fractions of ag-born and nonag-born workers who choose to work in agriculture are given by:

$$\begin{aligned} \phi_{A,t}(k) \; &\equiv \mathbb{P}_i \left[s_{i,t} = A \mid s_{i,t-1} = k \right] \\ &= \begin{cases} \dfrac{1}{1 + \left[\dfrac{B_{N,t}}{B_{A,t}} \cdot \dfrac{w_{N,t}}{w_{A,t}} \cdot \dfrac{(1-\tau)}{\kappa_A} \right]^{\gamma}} & \text{if} \quad k = A \\ \dfrac{1}{1 + \left[\dfrac{B_{N,t}}{B_{A,t}} \cdot \dfrac{w_{N,t}}{w_{A,t}} \cdot \kappa_N \right]^{\gamma}} & k = N \end{cases} \end{aligned} \tag{1}$$

where $k \in \{A, N\}$ indexes the worker's parent sector. Expressions analogous to (1) hold for the fractions of ag-born and nonag-born workers in nonagriculture, which also satisfy:

$$\phi_{N,t}(k) = 1 - \phi_{A,t}(k)$$

I visualize how inherited productivity and immobility wedges influence worker sorting with the numerical examples in Figure 4.

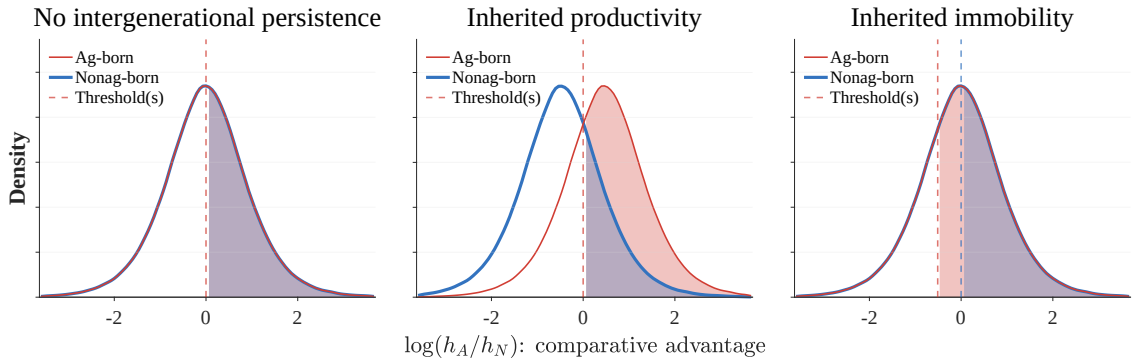


Figure 4: How Does Intergenerational Persistence Influence Sorting?

Notes: The red and blue curves depict the simulated distributions of log comparative advantage for ag-born and nonag-born workers, respectively, with each distribution normalized to integrate to one. Shading identifies the workers to the right of their group-specific dashed threshold who choose agriculture. The first panel sets $\kappa_A = \kappa_N = 1$ and $\tau = 0$, the second introduces inherited productivity through $\kappa_A, \kappa_N > 1$, and the third introduces inherited immobility through $\tau > 0$ while holding $\kappa_A = \kappa_N = 1$. Wages and amenities are held fixed across panels, and inherited immobility changes only the threshold faced by ag-born workers.

In this example, equally-sized ag-born and nonag-born subcohorts face relative wages and payoff shifters that, in the absence of intergenerational persistence, would lead to half of each subcohort choosing agriculture. This scenario is depicted in the first panel of Figure 4: the sorting of ag-born (red) and nonag-born (blue) workers into agriculture (denoted by the shaded area) is identical. In the second panel, I introduce inherited productivity through $\kappa_A, \kappa_N > 1$, which shifts the comparative advantage distribution of ag-born workers toward agriculture and that of nonag-born workers away from agriculture. Hence, the fraction of ag-born workers who choose agriculture increases, while the fraction of nonag-born workers who choose agriculture decreases. In the third panel, I introduce inherited immobility through $\tau > 0$ while holding $\kappa_A = \kappa_N = 1$. This shifts the threshold faced exclusively by ag-born workers toward lower comparative advantage in agriculture, which increases the fraction of ag-born workers who choose agriculture without affecting the fraction of nonag-born workers who choose agriculture.

Effective Labor Supplied by Ag-born and Nonag-born Workers The average effective labor supplied by k-born workers to sector j is given by:

$$h_{j,t}(k) \equiv \mathbb{E}_i \left[h_{i,j,t} \mid s_{i,t} = j,\ s_{i,t-1} = k \right] = \begin{cases} \kappa_j \cdot \bar{\gamma} \cdot \phi_{j,t}(k)^{-\frac{1}{\gamma}} & \text{if} \quad j = k \\ \bar{\gamma} \cdot \phi_{j,t}(k)^{-\frac{1}{\gamma}} & \quad j \neq k \end{cases} \tag{2}$$

where $\bar{\gamma} \equiv \Gamma\left(1 - \frac{1}{\gamma}\right)$ is a constant determined by the Gamma function $\Gamma(\cdot)$.

Two features of workers' average effective labor supply are worth highlighting. First, as in Lagakos and Waugh (2013), there are productivity gains from self-selection that are decreasing in the fraction of workers who sort into the sector.[14] Second, inherited productivity wedges *directly* increase the average effective labor supplied by k-born workers. Consider an economy with just the inherited productivity wedges $\kappa_A > 1$ and $\kappa_N > 1$. Figure 4 above illustrates that these wedges imply ag-born workers are more likely to choose agriculture than nonag-born workers. The direct influence of the inherited productivity wedge will raise the average skill (and moreover the average earnings) of ag-born workers in agriculture relative to nonag-born workers. However, nonag-born workers will benefit from higher gains from self-selection into agriculture than ag-born workers. In order to identify the inherited productivity wedges, I need to first adjust for the self-selection effect.

2.3. Aggregation

Equations (1) and (2) describe the sectoral allocations of workers and effective labor for ag-born and nonag-born workers, respectively. However, the aggregate allocations of workers and effective labor across sectors are the relevant objects for structural transformation. To aggregate across ag-born and nonag-born workers, I need to account for the composition of the cohort, which is determined by the aggregate worker allocation in the previous period.

[14] To illustrate this, consider the average skill supplied by nonag-born workers to agriculture if they were randomly assigned to sectors is given by $\bar{\gamma}$. Then, the productivity gains from self-selection are equal to $\phi_{A,t}(N)^{-\frac{1}{\gamma}} > 1$. Workers who sort into a sector are more productive, on average, than randomly assigned workers, but these gains from selection decline as more workers sort into the sector.

Aggregate Worker Allocations The fraction of workers in agriculture is given by:

$$\phi_{A,t} \quad = \quad \phi_{A,t-1} \cdot \phi_{A,t}(A) \quad + \quad (1 - \phi_{A,t-1}) \cdot \phi_{A,t}(N) \tag{3}$$

There is an analogous expression for the fraction of workers in nonagriculture, which moreover satisfies: $\phi_{N,t} = 1 - \phi_{A,t}$.

Effective Labor Supply The total effective labor supply in the agricultural sector is given by:

$$H_{A,t} \quad = \quad \phi_{A,t-1} \cdot \phi_{A,t}(A) \cdot h_{A,t}(A) \quad + \quad (1 - \phi_{A,t-1}) \cdot \phi_{A,t}(N) \cdot h_{A,t}(N) \tag{4}$$

There is an analogous expression for the total effective labor supply in nonagriculture, $H_{N,t}$.

Aggregate Income The total labor income in the economy is given by:

$$Y_t \quad = \quad w_{A,t} \cdot H_{A,t} \quad + \quad w_{N,t} \cdot H_{N,t} \tag{5}$$

2.4. Aggregate Responsiveness to Relative Wages

Already in the partial equilibrium environment, the model provides new insights into how intergenerational persistence shapes worker reallocation. Consider the relative allocation of k-born workers between nonagriculture and agriculture satisfies:

$$\frac{\partial \log \left(\phi_{N,t}(k) / \phi_{A,t}(k) \right)}{\partial \log \left(w_{N,t} / w_{A,t} \right)} \quad = \quad \gamma$$

Given a marginal 1-percent increase in the relative wage, k-born workers will experience a γ-percent increase in their relative allocation to nonagriculture. Refer to $\mathcal{E}(k) = \gamma$ as the within-subgroup elasticity of workers' relative sectoral allocation with respect to the relative wage.

A key implication of the model is that, even when both ag-born and nonag-born workers share the same within-subgroup elasticity $\mathcal{E}(A) = \mathcal{E}(N) = \gamma$, the aggregate elasticity will differ. Let $D_{i,t} \equiv \mathbb{1}\{s_{i,t} = A\}$ be an indicator for whether worker i chooses agriculture in period t. Then, the aggregate elasticity of workers' relative sectoral allocation with

respect to the relative wage is given by:

$$\mathcal{E} \equiv \frac{\partial \log\left(\phi_{N,t}/\phi_{A,t}\right)}{\partial \log\left(w_{N,t}/w_{A,t}\right)} = \gamma \cdot \left[1 - \frac{\text{Var}\left(\mathbb{E}[D_{i,t} \mid k]\right)}{\text{Var}(D_{i,t})}\right]$$

where $\text{Var}\left(\mathbb{E}[D_{i,t} \mid k]\right)$ is the variance of the expected sectoral choice between ag-born and nonag-born workers and $\text{Var}(D_{i,t})$ is the variance of sectoral choice across all workers.

Proposition 1. *Holding parental composition and the relative payoff shifter fixed, the elasticity of aggregate relative sectoral allocation satisfies:*

$$0 < \mathcal{E} \leq \mathcal{E}(k) = \gamma$$

The inequality is strict when both parental groups are present and their sector-choice probabilities differ; otherwise, $\mathcal{E} = \gamma$.

The attenuation term in the aggregate elasticity depends on how much sector choice variance is attributable to differences in sector choice between (rather than within) ag-born and nonag-born workers. In my setting, this attenuation term is given by:

$$\frac{\text{Var}\left(\mathbb{E}[D_{i,t} \mid k]\right)}{\text{Var}(D_{i,t})} = \frac{\phi_{A,t-1} \cdot (1 - \phi_{A,t-1}) \cdot [\phi_{A,t}(A) - \phi_{A,t}(N)]^2}{\phi_{A,t} \cdot (1 - \phi_{A,t})}$$

Hence, the aggregate elasticity is attenuated more when the economy is characterized by higher intergenerational persistence. Applying this expression to the observed parental composition and subgroup sector choices in the linked 1880 U.S. sample yields an aggregate elasticity 48 percent below the common within-group elasticity.

This distinction matters when using observed reallocation to discipline a Roy model. A researcher who knows γ but equates it with aggregate wage responsiveness will overstate the cohort's response when subgroups sort differently. Conversely, a researcher who consistently estimates aggregate wage responsiveness but interprets that estimate as γ will infer a shape parameter that is too low and therefore skill dispersion that is too high. The problem is not the estimate of aggregate responsiveness, but its interpretation through a model that omits subgroup heterogeneity. Parental sector is a particularly important source of subgroup heterogeneity in this setting. The same aggregation logic applies to other Roy-Fréchet applications in which groups defined by, e.g., gender, race, or nativity face different sectoral incentives. Attenuation does not require different shape parameters across groups: a common γ and different sector choice probabilities are sufficient.

In the next subsection, I discuss how to identify γ from the conditional variance of log earnings within subgroups. This avoids the bias that arises from estimating the aggregate elasticity and interpreting it as γ.

2.5. Identification

There are four labor supply parameters to identify: the Fréchet shape parameter $\gamma > 1$, the inherited productivity wedges $\kappa_A \geq 1$ and $\kappa_N \geq 1$, and the inherited immobility wedge $\tau \in [0, 1)$. I exploit the structure of the model to identify these parameters from four moments concerning workers' income and sectoral allocations.

Fréchet Shape Parameter, $\gamma > 1$ Define the maximum indirect utility of a k-born worker i in their optimally chosen sector as $V_{i,t}^{\max}(k) \equiv \max_{j \in \{A,N\}} V_{i,j,t}$. This object is also Fréchet distributed with shape parameter γ, but with scale parameter $\Lambda_t(k)$, which is given by:

$$\Lambda_t(k) \equiv \begin{cases} \left[(B_{A,t}\kappa_A w_{A,t})^\gamma + (B_{N,t}(1-\tau)w_{N,t})^\gamma \right]^{\frac{1}{\gamma}} & \text{if} \quad k = A \\ \left[(B_{A,t} w_{A,t})^\gamma + (B_{N,t}\kappa_N w_{N,t})^\gamma \right]^{\frac{1}{\gamma}} & \quad k = N \end{cases}$$

A generic property of Fréchet distributed random variables is that they can be re-written in terms of a standard Gumbel distributed random variable:

$$\log V_{i,t}^{\max}(k) = \log \Lambda_t(k) + \frac{1}{\gamma} u_{i,t}(k)$$

where $u_{i,t}(k)$ is Gumbel distributed with location parameter 0 and scale parameter 1. As shown in Appendix A.2.1, the maximum indirect utility is independent of the identity of the maximizing sector, so this distribution is unchanged after conditioning on sector choice.

Recall, each worker's income is equal to the skill they supply to their chosen sector multiplied by the wage per efficiency unit in that sector, $y_{i,j,t} \equiv w_{j,t} \cdot h_{i,j,t}$. Then, each k-born worker who optimally chooses sector j has realized labor income given by:

$$y_{i,j,t}(k) \equiv \begin{cases} \frac{V_{i,t}^{\max}(k)}{B_{j,t}(1-\tau)} & \text{if} \quad j = N,\ k = A \\ \frac{V_{i,t}^{\max}(k)}{B_{j,t}} & \text{otherwise} \end{cases}$$

As such, logged individual income satisfies:

$$\log y_{i,j,t}(k) \quad = \quad \log \Lambda_t(k) \quad - \quad \log B_{j,t} \quad - \quad \mathbb{1}_i \left\{ s_{i,t} = N,\ s_{i,t-1} = A \right\} \cdot \log(1-\tau) \quad + \quad \frac{1}{\gamma} u_{i,t}(k)$$

Since the variance of a standard Gumbel distributed random variable is $\pi^2/6$, then the Fréchet skill dispersion parameter γ is identified from the variance of logged individual income, conditional on sectoral choice and parental sector:

$$\text{Var}\left[\log y_{i,j,t}(k) \mid s_{i,t} = j,\ s_{i,t-1} = k\right] \quad = \quad \frac{\pi^2}{6} \cdot \frac{1}{\gamma^2} \tag{ID-1}$$

Inherited Productivity Wedges, $\kappa_A \geq 1$ and $\kappa_N \geq 1$ Since each workers' labor income satisfies $y_{i,j,t} = w_{j,t} \cdot h_{i,j,t}$, the average labor income of k-born workers who choose sector j is proportional to their average effective labor supply:

$$\begin{aligned} y_{j,t}(k) \quad &\equiv \quad \mathbb{E}_i \left[y_{i,j,t}(k) \mid s_{i,t} = j,\ s_{i,t-1} = k \right] \\ &= \quad w_{j,t} \cdot \begin{cases} \kappa_j \cdot \bar{\gamma} \cdot \phi_{j,t}(k)^{-\frac{1}{\gamma}} & \text{if} \qquad k = j \\ \bar{\gamma} \cdot \phi_{j,t}(k)^{-\frac{1}{\gamma}} & \qquad\quad k \neq j \end{cases} \end{aligned}$$

The average income ratio between ag-born and nonag-born workers in agriculture is then given by:

$$\frac{y_{A,t}(A)}{y_{A,t}(N)} \quad = \quad \kappa_A \ \cdot \ \left(\frac{\phi_{A,t}(A)}{\phi_{A,t}(N)} \right)^{-\frac{1}{\gamma}} \tag{ID-2}$$

which is the product of the inherited productivity wedge and the relative gains from self-selection into agriculture. As noted above, the gains from selection into agriculture will typically be larger for nonag-born workers than ag-born workers because the former are less likely to choose agriculture. The inherited productivity wedge κ_A is identified from the average income ratio between ag-born and nonag-born workers in agriculture after adjusting for the relative gains from self-selection. Likewise, the inherited productivity wedge κ_N is identified from:

$$\frac{y_{N,t}(A)}{y_{N,t}(N)} \quad = \quad \frac{1}{\kappa_N} \ \cdot \ \left(\frac{\phi_{N,t}(A)}{\phi_{N,t}(N)} \right)^{-\frac{1}{\gamma}} \tag{ID-3}$$

Inherited Immobility Wedge, $\tau \in [0, 1)$ One measure of an economy's intergenerational persistence in sector choice is the odds ratio, which represents the relative odds that an

ag-born worker chooses to work in agriculture compared to a nonag-born worker:

$$\mathcal{O} \equiv \frac{\phi_{A,t}(A)/\phi_{N,t}(A)}{\phi_{A,t}(N)/\phi_{N,t}(N)} = \frac{\left[\frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \kappa_N\right]^\gamma}{\left[\frac{B_{N,t}}{B_{A,t}} \cdot \frac{w_{N,t}}{w_{A,t}} \cdot \frac{(1-\tau)}{\kappa_A}\right]^\gamma}$$

Then, conditional on the inferred Fréchet shape parameter γ and inherited productivity wedges κ_A and κ_N, the inherited immobility wedge τ is identified as the economy's only remaining source of persistence:

$$\mathcal{O} = \left(\frac{\kappa_A \cdot \kappa_N}{1-\tau}\right)^\gamma \tag{ID-4}$$

2.6. Estimation

To estimate these parameters, I again rely on the historical census data used in section 1 to establish the paper's motivating facts. That is, I focus on the linked sample of 20–29 year-old U.S.-born men working in 1940 linked to their working fathers in the prior census. Moreover, I restrict attention to individuals with positive wage and salary income, positive weeks worked, and at least 35 hours per week worked. In the appendix, I report estimates using annual or weekly earnings, which are broadly similar to my baseline estimates using hourly earnings.

There are two noteworthy limitations of these data. First, the census topcodes income at \$5,000 in 1940. Topcoded workers constitute 0.1 percent of the estimation sample, and their capped earnings constitute 0.5 percent of its total wage and salary income. Topcoding compresses measured log-income dispersion and may therefore lead to slight overestimates of the Fréchet shape parameter γ. Moreover, this would also lead to slight underestimates of the inherited productivity wedges κ_A and κ_N because I would attribute too much of the average income gaps to the relative gains from self-selection. As a robustness check, I multiply every topcoded observation by 2.926 before constructing hourly earnings. This factor makes the affected observations account for 1.5 percent of total earnings, matching the 1940 salary income share of the top 0.1 percent reported by Atkinson, Piketty and Saez (2011).

Second, with only wage and salary income reported in the census, I do not observe self-employed farming or business income.[15] As such, in the linked sample I only observe

[15] Carruthers and Wanamaker (2017) discuss the analogous limitation created by the absence of self-

income for about 46 percent of agricultural workers and 90 percent of nonagricultural workers. This is a serious limitation. Suppose, for example, that the most productive ag-born workers are more likely to sort into self-employed farming rather than wage work while nonag-born workers have no selection into self-employment. As such, the variance of observed earnings will be smaller than the population variance of earnings, which would lead to an upward bias in the estimate of γ. In addition, the inherited productivity wedge in agriculture, κ_A, would be underestimated primarily because the observed ag-born wage workers are less productive on average than the unobserved self-employed ag-born workers. The inherited immobility wedge τ would then be overestimated.

To proceed, I assume that, conditional on destination sector and parental sector, selection into the earnings sample is independent of latent sectoral productivity. Maintaining this assumption is needed to rule out any bias in the estimates of γ, κ_A, κ_N, and τ that would arise from selection into self-employment. It also encompasses selection induced by the restrictions on positive earnings, weeks worked, and usual weekly hours. Under this assumption, the earnings moments in the restricted sample represent the corresponding moments among all workers in each sector-by-parental-sector cell.

The observed income data may reflect other sources of heterogeneity that the model abstracts from including age and local labor market conditions. Indeed, the motivating facts in section 1 show that local labor markets play a small but non-trivial role in explaining intergenerational persistence in sector choice. My baseline parameter estimates are based on the identification strategy consistent with the model as outlined above; however, I report robustness checks that condition on these additional sources of heterogeneity. The results are broadly similar to the baseline estimates.

Estimates of γ As outlined above, the Fréchet shape parameter γ is identified from the variance of logged individual income, conditional on sectoral choice and parental sector. I report my preferred hourly estimate of γ in Table 2, together with estimates that add controls or adjust topcoded earnings. The baseline variance-based estimate is 2.076, while adding fixed effects for age and adulthood commuting zone raises the estimate to 2.288. Rescaling topcoded earnings yields 2.063, very close to the baseline estimate. Appendix Table 13 reports corresponding estimates using annual and weekly earnings. These estimates lie within the range of Fréchet shape parameters in related Roy models inferred from wage dispersion Kuralbayeva and Stefanski (2013), maximum-likelihood

employment income in the 1940 Census and likewise exclude individuals without recorded earnings from their baseline analysis.

fits of residual earnings distributions Hsieh et al. (2019), and farm-size distributions Eckert and Peters (2022). Appendix A.2 reports results for alternative earnings measures and sensitivity to the topcode adjustment.

Table 2: Estimates of the Fréchet Shape Parameter: Hourly Earnings

	Baseline	Added FE	Topcode adjusted
$\widehat{\gamma}$	2.076	2.288	2.063
Observations	2,077,253	2,077,253	2,077,253
Quantile RMSE (1st–99th percentiles)	0.337	0.307	0.340

Note: Columns 1 and 2 infer γ from the residual variance of log hourly earnings; column 2 additionally controls for adult commuting zone and age. Column 3 replaces each topcoded income with 2.926 times the censoring threshold, calibrated to match a 1.452 percent top-0.1-percent income share. All columns use 2,077,253 observations with positive earnings and weeks and at least 35 weekly hours; no log-earnings restriction is imposed. Quantile RMSE measures fit across the 1st–99th residual percentiles in log points; lower values indicate better fit.

The variance-based estimates are directly aimed at recovering the Fréchet shape parameter. In contrast, according to Proposition 1, an estimator targeting the aggregate elasticity of worker reallocation would recover a coefficient substantially below these γ estimates. I present preliminary evidence of this in Appendix Table 11, which reports estimates of the aggregate elasticity of worker reallocation using commuting-zone-by-industry variation in relative wages and sectoral allocations between 1940 and 1950. The population-weighted aggregate reallocation elasticity is 1.067, which is 48 percent below the baseline variance-based estimate of γ. This illustrates that studies exploiting aggregate reallocation to infer the Fréchet shape parameter may substantially overestimate underlying skill dispersion if they ignore sources of subgroup heterogeneity in sector choice such as parental sector.

Estimates of Inherited Productivity and Immobility Wedges Next, I estimate the inherited productivity wedges κ_A and κ_N from average income differences between ag-born and nonag-born workers employed in the same sector net of the gains from self-selection. Then, I estimate the inherited immobility wedge τ from the remaining difference between the sector choices of ag-born and nonag-born workers.

Let $\widehat{y}_j(k)$ denote mean income in the relevant income sample among workers employed in sector j whose fathers worked in sector k, and let $\widehat{\phi}_j(k)$ denote the corresponding sector-choice probability. Given the estimated $\widehat{\gamma}$, I compute the inherited productivity

wedges as:

$$\widehat{\kappa}_A = \frac{\widehat{y}_A(A)/\widehat{\phi}_A(A)^{-\frac{1}{\widehat{\gamma}}}}{\widehat{y}_A(N)/\widehat{\phi}_A(N)^{-\frac{1}{\widehat{\gamma}}}} \quad \text{and} \quad \widehat{\kappa}_N = \frac{\widehat{y}_N(N)/\widehat{\phi}_N(N)^{-\frac{1}{\widehat{\gamma}}}}{\widehat{y}_N(A)/\widehat{\phi}_N(A)^{-\frac{1}{\widehat{\gamma}}}}$$

Hence, the inherited productivity is identified by the ratio of average income between ag-born and nonag-born workers in the same sector after removing the gains from self-selection.

For example, the average hourly earnings of ag-born workers in agriculture is 0.16 and for nonag-born workers in agriculture is 0.34. The gains from self-selection play a critical role here. For ag-born workers, their average income in agriculture is roughly 35% higher due to gains from self-selection. However, for nonag-born workers, their average income in agriculture is roughly 312% higher due to gains from self-selection. After adjusting for these selection effects, the average income of ag-born workers in agriculture is roughly 42% higher than that of nonag-born workers in agriculture, which is attributed to the inherited productivity wedge, $\widehat{\kappa}_A$. Similarly, the selection-adjusted average income of nonag-born workers in nonagriculture is roughly 81% higher than that of ag-born workers in nonagriculture, which is attributed to the inherited productivity wedge, $\widehat{\kappa}_N$.

A critical assumption I leverage to identify and estimate these inherited productivity wedges is that both ag-born and nonag-born workers face the same wage per efficiency unit in each sector. There are at least two reasons to doubt this assumption. First, ag-born and nonag-born workers may live in systematically different locations where sectoral wages differ. My specifications that control for adult commuting zone fixed effects explicitly address this concern. Second, workers may benefit from more favorable wage offers in their parents' sectors due to discrimination or nepotism. By limiting my attention to workers who moved across commuting zones between childhood and adulthood, I can partially evaluate the plausibility of this concern.[16] I present the results of this exercise in Appendix Table 9. Though attenuated, I show that substantial inherited productivity wedges persist even among workers who move across commuting zones, which makes it less likely that the observed earnings gaps arise entirely from local employers conditioning wage offers on workers' family backgrounds. Moving forward, I proceed under a maintained assumption that ag-born and nonag-born workers face the same wage per efficiency unit in each sector, which allows me to interpret the estimated earnings gaps as inherited productivity wedges.

[16] Of course, there is selection into moving, which may bias these estimates. I view this exercise as a plausibility check rather than a formal test of the maintained assumption.

The inherited immobility wedge is identified as the remaining source of intergenerational persistence in sector choice not explained by inherited productivity wedges:

$$\widehat{\tau} \quad = \quad 1 - \frac{\widehat{\kappa}_A \widehat{\kappa}_N}{\widehat{\mathcal{O}}^{1/\widehat{\gamma}}}.$$

where $\widehat{\mathcal{O}}$ is the observed odds ratio of ag-born workers choosing agriculture relative to nonag-born workers. Using sector choices from the full linked sample to estimate the inherited immobility wedge, I find that ag-born workers sort as if they value nonagricultural work about 33 percent less than an equally-skilled nonag-born worker. Expressed as shares of observed log-odds persistence, inherited productivity accounts for 72.7 percent and inherited immobility for 27.3 percent; Appendix A.2.2 derives this decomposition. Using the topcode-adjusted estimate of γ yields nearly identical estimates: $\widehat{\kappa}_A = 1.615$, $\widehat{\kappa}_N = 1.801$, and $\widehat{\tau} = 0.328$. I report the estimates of κ_A, κ_N, and τ in Table 3.

Table 3: Inherited Productivity and Immobility Wedges: Hourly Earnings

	Baseline	Added FE	Topcode adjusted
Panel A: Inferred wedges			
Inherited ag. productivity, $\widehat{\kappa}_A$	1.604	1.323	1.615
Inherited nonag. productivity, $\widehat{\kappa}_N$	1.797	1.437	1.801
Inherited mobility wedge, $\widehat{\tau}$	0.328	0.281	0.328
Panel B: Mean hourly earnings			
Agriculture, ag-born: $\widehat{y}_A(A)$	0.19	0.20	0.19
Agriculture, nonag-born: $\widehat{y}_A(N)$	0.37	0.32	0.37
Nonagriculture, ag-born: $\widehat{y}_N(A)$	0.47	0.52	0.47
Nonagriculture, nonag-born: $\widehat{y}_N(N)$	0.60	0.59	0.60
Panel C: Intergenerational persistence			
Ag. employment share, ag-born	0.533	0.472	0.533
Ag. employment share, nonag-born	0.053	0.088	0.053
Sector-choice odds ratio	20.541	9.266	20.541

Note: The table reports inherited productivity premia, $\widehat{\kappa}_A$ and $\widehat{\kappa}_N$, and the inherited immobility wedge, $\widehat{\tau}$, using hourly earnings from the 1940 linked census. Columns 1–3 use the baseline, added-fixed-effect, and topcode-adjusted estimates of γ from Table 2; column 2 also adjusts the underlying moments for adult commuting zone and age. Earnings moments entering κ_j use the income sample defined in Table 2; all sector-choice moments use the full sample of workers with valid worker and parental sectors. Panel B reports the income-sample means, while Panel C reports the full-sample sector choices also used to estimate κ_j and τ. Column 3 holds the baseline moments fixed and changes only $\widehat{\gamma}$.

Discussion The motivating facts established in section 1 show that ag-born and nonag-born workers differ in their sector choices and average earnings. In order to provide a transparent interpretation of these differences, I leverage the structure of this partial equilibrium Roy-type model. Taking the model to data, my estimates suggest that ag-born workers are disproportionately drawn to agriculture both because they are more productive in agriculture and because they find nonagricultural work less appealing or more difficult to pursue than nonag-born workers. Related structural estimates in other settings also imply substantial inherited productive advantages and nonpecuniary attachments, although the magnitudes vary considerably (Cassan, Keniston and Kleineberg, 2021; Almgren, Kramer and Sigurdsson, 2025).

A critical assumption throughout this analysis is that workers draw their skills from the same Fréchet distribution. This assumption permits the tractable closed-form solutions that I exploit to identify the labor supply parameters and also to derive analytical insights here and in the subsequent general equilibrium analysis. Appendix Table 10 reports SMM estimates under two alternative distributions: bivariate Fréchet and bivariate lognormal. These specifications allow for the possibility of different skill dispersion in each sector and that individual workers' skills in agriculture and nonagriculture are correlated. The bivariate Fréchet yields much higher estimates of κ_A and slightly lower estimates of κ_N and τ than the baseline specification, while the bivariate lognormal yields much lower estimates of κ_A and κ_N with much higher estimates of τ. Because these alternative distributions imply materially different inherited wedges, I later evaluate the quantitative counterfactuals using each alternative wedge vector.

Beyond the identification and estimation of the model's labor supply parameters, I also established a key insight into how intergenerational persistence shapes structural transformation. Proposition 1 shows that the aggregate elasticity of worker reallocation is attenuated by intergenerational persistence in sector choice. In particular, workers may be only half as responsive to changes in relative wages in 1880 as they would be in a counterfactual economy without intergenerational persistence. This insight is relevant to the broad array of research exploiting Roy frameworks to model labor allocation, especially in quantitative macroeconomic and trade. In the next section, I embed the labor supply block into a general equilibrium environment in order to trace these labor demand shocks from their origins through the economy's structural transformation. I show that the inherited productivity and immobility wedges dampen the newest generation's general equilibrium response to labor demand shocks to different degrees. Moreover, I show that intergenerational persistence provides a dynamic transmission mechanism through

which the next generation's labor supply responds to the current generation's reallocation.

3. General Equilibrium Analysis

Here I describe the general equilibrium closure of the model, especially the production and consumption side of the economy. Then, I define the competitive equilibrium and characterize how intergenerational persistence alters the economy's response to the drivers of structural transformation.

3.1. Production and Consumption

The economy features representative firms in each sector. Sectoral goods are allocated through an aggregate consumption system with the nonhomothetic CES aggregator below.

Sectoral Production Each sector has a representative firm that produces output using labor as the only factor input according to:

$$Q_{A,t} \;=\; Z_{A,t} \cdot H_{A,t} \qquad \text{and} \qquad Q_{N,t} \;=\; Z_{N,t} \cdot H_{N,t} \tag{6}$$

where $Z_{j,t} > 0$ is the production technology of sector j.

Labor Demand The firm takes the price of goods, $p_{j,t}$, and the wage per efficiency unit of labor, $w_{j,t}$, as given, and chooses labor demand to maximize profits:

$$\max_{H_{j,t}} \; p_{j,t} Q_{j,t} - w_{j,t} H_{j,t}$$

which implies that the wage per efficiency unit of labor is equal to the value of the marginal product of labor:

$$w_{A,t} \;=\; p_{A,t} \cdot Z_{A,t} \qquad \text{and} \qquad w_{N,t} \;=\; p_{N,t} \cdot Z_{N,t} \tag{7}$$

Consumption Let $e(p_t, Q_t)$ denote the minimum expenditure on sectoral goods required to attain consumption index Q_t determined by the nonhomothetic CES production func-

tion described in Comin, Lashkari and Mestieri (2021), which extends the demand-based structural-transformation mechanisms in Kongsamut, Rebelo and Xie (2001) and Boppart (2014):

$$\sum_{j\in\{A,N\}} \Omega_j^{1/\sigma} \left(\frac{X_{j,t}}{Q_t^{\vartheta_j}}\right)^{\frac{\sigma-1}{\sigma}} = 1, \tag{8}$$

where $X_{j,t}$ is the quantity of sector j goods demanded, $\Omega_j > 0$ is the demand weight on sector j, $\sigma \in (0,1)$, and $\vartheta_N > \vartheta_A = 1$. If $\vartheta_A = \vartheta_N = 1$, then this recovers the typical homothetic CES aggregator with elasticity of substitution σ.

Define the sector j output share as $\omega_{j,t} \equiv p_{j,t}X_{j,t}/(P_tQ_t)$, where P_t is the expenditure per unit of aggregate consumption:

$$P_t = \frac{Y_t}{Q_t} \tag{9}$$

Then, relative quantity demand satisfies:

$$\frac{X_{A,t}}{X_{N,t}} = \frac{\Omega_A}{\Omega_N}\left(\frac{p_{N,t}}{p_{A,t}}\right)^{\sigma}\left(\frac{Y_t}{P_t}\right)^{(\vartheta_A-\vartheta_N)(1-\sigma)}. \tag{10}$$

Multiplying relative quantity demand by $p_{A,t}/p_{N,t}$ gives relative output shares:

$$\frac{\omega_{A,t}}{\omega_{N,t}} = \frac{\Omega_A}{\Omega_N}\left(\frac{p_{N,t}}{p_{A,t}}\right)^{\sigma-1}\left(\frac{Y_t}{P_t}\right)^{(\vartheta_A-\vartheta_N)(1-\sigma)}. \tag{11}$$

3.2. Competitive Equilibrium

Let $p_{A,t} = 1$ be the numeraire and $p_t \equiv p_{N,t}/p_{At}$ be the relative price of nonagricultural goods. For an initial worker allocation $\phi_{A,0}$ and the exogenous fundamentals

$$\{Z_{j,t}, B_{j,t}\}_{j\in\{A,N\},t>0}$$

a competitive equilibrium is a sequence of allocations

$$\{\phi_{j,t}(k), h_{j,t}(k), \phi_{j,t}, H_{j,t}, Y_t, Q_{j,t}, X_{j,t}, \omega_{j,t}, Q_t\}_{j,k\in\{A,N\},t>0}$$

and prices

$$\{w_{j,t}, p_t, P_t\}_{j\in\{A,N\},t>0}$$

such that:

1. (Non)ag-born workers optimally choose sectors according to equation (1) such that the aggregate allocation satisfies equation (3).
2. The average effective labor supplied to each sector by ag-born and nonag-born workers satisfies equation (2), and the total effective labor supplied to each sector satisfies equation (4).
3. Aggregate labor income satisfies equation (5).
4. Competitive sectoral firms produce according to equation (6) and pay each efficiency unit its value marginal product according to equation (7).
5. The aggregate consumption system features sectoral good demand satisfying equation (11) and (10) such that aggregate consumption satisfies equation (8), and the expenditure index satisfies equation (9).
6. Sectoral goods markets clear:

$$X_{j,t} = Q_{j,t} \qquad \text{for each} \qquad j \in \{A, N\}. \tag{12}$$

given the parameters $\{\gamma, \kappa_A, \kappa_N, \tau, \sigma, \vartheta_A, \vartheta_N, \Omega_A, \Omega_N\}$.

3.3. Characterizing the Influence of Intergenerational Persistence

In this model, all worker reallocation results from either a labor demand shock originating in sectoral technologies (i.e., $Z_{j,t}$) or a labor supply shock originating in the payoff shifter (i.e., $B_{j,t}$). For expositional simplicity, I focus on a 1-percent permanent improvement in the agricultural technology. To explore how intergenerational persistence shapes the economy's response to this shock, I simplify the transmission into three components depicted in the figure below.

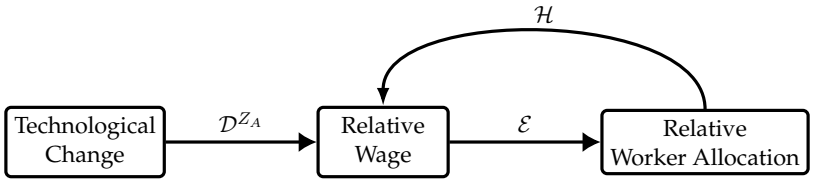


Figure 5: Diagram of the Contemporaneous Transmission of Technological Change

Notes: $\mathcal{D}^{Z_A}$ is the technology-induced shift in relative labor demand, $\mathcal{E}$ is the elasticity of relative worker allocation to the relative wage, and $\mathcal{H}$ is the relative-wage feedback from worker reallocation.

Consistent with this diagram, I reduce the equilibrium system to two local equations: a relative headcount labor supply schedule and an inverse relative labor demand schedule expressed in terms of headcount allocations.[17] The relative labor supply schedule is:

$$d\log\frac{\phi_{N,t}}{\phi_{A,t}} \quad = \quad \mathcal{E}\, d\log\frac{w_{N,t}}{w_{A,t}},$$

where $\mathcal{E}$ is the aggregate worker reallocation elasticity presented in the previous section. The inverse relative labor-demand schedule is

$$d\log\frac{w_{N,t}}{w_{A,t}} \quad = \quad \mathcal{D}^{Z_A}\, d\log Z_{A,t} \quad + \quad \mathcal{H}\, d\log\frac{\phi_{N,t}}{\phi_{A,t}}.$$

where $\mathcal{D}^{Z_A}$ is the shift in relative labor demand from the agricultural technology shock and $\mathcal{H}$ is the feedback of worker reallocation into the relative wage. Appendix **??** formally defines these objects and derives both schedules from the full equilibrium system.

Solving the two schedules gives the immediate general equilibrium response of the relative worker allocation:

$$d\log\frac{\phi_{N,t}}{\phi_{A,t}} \quad = \quad \frac{\mathcal{E}}{1 \;-\; \mathcal{E}\mathcal{H}}\, \mathcal{D}^{Z_A}\, d\log Z_{A,t}.$$

Because $d\phi_{A,t} = -\phi_{A,t}(1-\phi_{A,t})\, d\log(\phi_{N,t}/\phi_{A,t})$, the contemporaneous reallocation of agricultural workers is therefore

$$\frac{d\phi_{A,t}}{d\log Z_{A,t}} \quad = \quad -\phi_{A,t}(1-\phi_{A,t})\, \frac{\mathcal{E}}{1 \;-\; \mathcal{E}\mathcal{H}}\, \mathcal{D}^{Z_A}.$$

I use this decomposition to compare contemporaneous responses across economies with different sources of intergenerational persistence.

Stylized Economies Refer to the first economy as the "benchmark" economy, which has no intergenerational persistence (i.e., $\kappa_A = \kappa_N = 1$ and $\tau = 0$). Refer to the second economy as the "IGP-κ" economy, which has intergenerational persistence from a symmetric inherited productivity only (i.e., $\kappa_A = \kappa_N = \kappa > 1$, and $\tau = 0$). Refer to the third economy as the "IGP-τ" economy, which has intergenerational persistence from inherited immo-

[17] I distinguish headcount allocations from effective labor allocations because the former are the objects of interest in this paper, while the latter are the objects that enter the production system. Effective labor allocations not only account for the number of workers in each sector, but also for the average effective labor supplied by those workers.

bility only (i.e., $\kappa_A = \kappa_N = 1$ and $\tau > 0$). I restrict the two IGP economies to have the same gap in the allocations of ag-born and nonag-born workers:

$$\phi_{A,t}(A)^{\text{IGP-}\kappa} - \phi_{A,t}(N)^{\text{IGP-}\kappa} \quad = \quad \phi_{A,t}(A)^{\text{IGP-}\tau} - \phi_{A,t}(N)^{\text{IGP-}\tau} \quad > \quad 0$$

Finally, I require that all three economies have the same stationary allocation of workers absent any technological change, $\phi_{A,t} = \phi_{A,t-1}$.

How Workers Respond to Wages in Partial Equilibrium, $\mathcal{E}$ The main channel through which intergenerational persistence shapes the economy's immediate response to technological change is through the elasticity of worker allocation with respect to the relative wage, $\mathcal{E}$. In the previous section, I showed that intergenerational persistence reduces the responsiveness of workers to changes in relative wages such that $\mathcal{E} < \gamma$. Under the additional restrictions imposed on the three stylized economies, I note:

$$0 \quad < \quad \mathcal{E}^{\text{IGP-}\kappa} \quad = \quad \mathcal{E}^{\text{IGP-}\tau} \quad < \quad \mathcal{E}^{\text{bench}} \quad = \quad \gamma$$

That is, the aggregate elasticity of worker reallocation is strictly smaller in the presence of intergenerational persistence regardless of whether that persistence arises from inherited productivity or inherited immobility. This plays a critical role but it does not fully characterize how intergenerational persistence affects the contemporaneous response of workers to technological change in general equilibrium. The full characterization also depends on the relative-wage feedback from worker reallocation, $\mathcal{H}$, which does depend on the source of intergenerational persistence.

How Workers Respond to Technical Change in General Equilibrium In order to provide a transparent characterization of how intergenerational persistence shapes workers' contemporaneous response to technological change, I restrict the production-side of the economy to be homothetic (i.e., $\vartheta_A = \vartheta_N = 1$). This shuts down the aggregate income channel through which changes in effective labor supply feed back into the relative wage, but preserves the relative price feedback channel. With this restriction, the general equilibrium feedback of worker reallocation into the relative wage is greater when the economy features intergenerational persistence, especially from inherited immobility. Formally:

Lemma 1. *The magnitude of the general equilibrium feedback of worker reallocation into the*

relative wage is strictly larger in the presence of intergenerational persistence, and it is largest when that persistence arises from inherited immobility.

$$\mathcal{H}^{\textit{IGP-}\tau} \quad < \quad \mathcal{H}^{\textit{IGP-}\kappa} \quad \leq \quad \mathcal{H}^{\textit{bench}} \quad = \quad -\frac{1}{\sigma} \cdot \frac{\gamma - 1}{\gamma}$$

where the equality holds only when $\phi_{A,t-1} = 0.5$.

Workers reallocating out of agriculture implies a reallocation of effective labor that induces a decrease in the relative wage. However, the magnitude of that general equilibrium feedback is larger in the presence of intergenerational persistence, especially due to inherited immobility. This together with the reduction in workers' responsiveness to wages, $\mathcal{E}$, implies that the contemporaneous response of workers to technological change is smaller in the presence of intergenerational persistence. In particular:

Proposition 2. *Improvements in agricultural technology induce a reallocation of workers out of agriculture, but the magnitude of that reallocation is smaller in the presence of intergenerational persistence.*

$$\frac{d\phi_{A,t}}{d\log Z_{A,t}}^{\textit{bench}} \quad < \quad \frac{d\phi_{A,t}}{d\log Z_{A,t}}^{\textit{IGP-}\kappa} \quad < \quad \frac{d\phi_{A,t}}{d\log Z_{A,t}}^{\textit{IGP-}\tau} \quad < \quad 0$$

In particular, workers reallocate less when the economy features inherited immobility than when it features inherited productivity.

Figure 6 illustrates the equilibrium forces underlying Proposition 2. Because the relative headcount labor supply schedule has slope $1/\mathcal{E}$, the smaller aggregate reallocation elasticity in the two IGP economies makes their supply schedules steeper than in the benchmark economy. Moreover, the source of persistence distinguishes their inverse labor demand schedules: the more negative $\mathcal{H}$ in the IGP-τ economy produces a steeper downward-sloping schedule than in the IGP-κ economy, whose schedule is in turn steeper than in the benchmark economy. Consequently, the same technology-induced shift in relative labor demand generates the largest increase in $\log(\phi_N/\phi_A)$ in the benchmark economy, a smaller increase under inherited productivity, and the smallest increase under inherited immobility. Thus intergenerational persistence dampens the immediate structural transformation accommodated by the newest generation due to both a weaker aggregate worker reallocation elasticity and a stronger relative wage feedback. Moreover, inherited immobility produces the greater immediate bottleneck to structural transformation due to its stronger general equilibrium feedback.

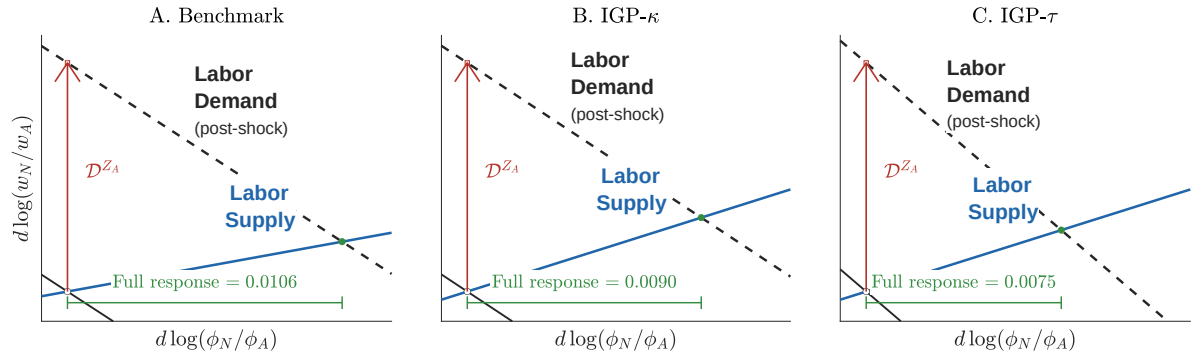


Figure 6: Headcount Relative Labor Supply and Demand with a Shock to Agricultural Technology

Notes: The figure plots log changes in the nonagricultural-to-agricultural employment ratio against log changes in the relative wage, normalizing the initial equilibrium to the origin. Blue lines show relative labor supply; solid and dashed black lines show inverse labor demand before and after a one-percent increase in agricultural technology. The red arrow is the demand shift, the green circle the new equilibrium, and the green bracket the resulting worker reallocation. All panels impose homothetic demand, $\gamma = 2.076$, $\sigma = 0.3$, and $\phi_A = 0.665$. Panel A has no intergenerational persistence; Panels B and C generate an odds ratio of 25 entirely through symmetric inherited productivity ($\kappa_A = \kappa_N = 2.171$) or inherited immobility ($\tau = 0.788$). These two economies have the same allocation gap but different general-equilibrium wage feedbacks.

Future Generations Respond to Past Reallocation Worker reallocation alters the composition of ag-born and nonag-born workers in subsequent generations. In particular, if workers are induced to leave agriculture in the current generation, then there are fewer ag-born workers in the next generation. A direct consequence of this compositional change, holding worker's choices fixed, is that the next generation allocates fewer workers to agriculture. An indirect consequence is that, still holding worker's choices fixed, the next generation's effective labor supply endogenously shifts such that relative wages change and induce further worker reallocation. Define these two shifts as:

$$\mathcal{F}^{direct} \equiv \left.\frac{\partial \phi_{A,t+1}}{\partial \phi_{A,t}}\right|_{d\phi_{j,t+1}(k)=0} \quad \text{and} \quad \mathcal{F}^{indirect} \equiv \left.\frac{\partial \log(w_{N,t+1}/w_{A,t+1})}{\partial \phi_{A,t}}\right|_{d\phi_{j,t+1}(k)=0}.$$

Combining these two shifts with the aggregate worker reallocation elasticity and relative wage feedback characterized above gives the total dynamic response: [18]

$$\frac{d\phi_{A,t+1}}{d\phi_{A,t}} = \mathcal{F}^{direct} - \phi_{A,t+1}\phi_{N,t+1} \cdot \frac{\mathcal{E}}{1-\mathcal{E}\mathcal{H}} \cdot \mathcal{F}^{indirect}$$

[18] Appendix **??** derives this expression from the full period-$t+1$ equilibrium system.

First, the composition of ag-born and nonag-born workers directly affects the aggregate worker allocation through $\mathcal{F}^{direct}$. Ag-born workers are more likely to work in agriculture than nonag-born workers, so a smaller fraction of ag-born workers implies a smaller fraction of workers in agriculture. In particular, holding wages fixed, workers respond directly to past reallocation according to:

$$\mathcal{F}^{direct} \quad = \quad \phi_{A,t+1}(A) - \phi_{A,t+1}(N)$$

This direct transmission of past worker reallocation is nonzero only when the economy features intergenerational persistence, though it is independent of the source of that persistence. Formally:

Lemma 2. *Holding wages fixed, future generations respond to past reallocation according to:*

$$0 \quad = \quad \mathcal{F}^{direct,bench} \quad < \quad \mathcal{F}^{direct,IGP\text{-}\kappa} \quad = \quad \mathcal{F}^{direct,IGP\text{-}\tau}$$

Hence, intergenerational persistence directly propagates past reallocation forward to future generations.

Second, the composition of ag-born and nonag-born workers also affects the effective labor supplied to each sector through equation (4). As such, a reduction in ag-born workers implies a decrease (increase) in the effective labor supplied to (non)agriculture in the next generation, all else equal. This moreover induces a change in the relative wage, which leads to a reallocation of workers. By restricting the production side of the economy to be homothetic, I provide a transparent characterization of how intergenerational persistence shapes the indirect transmission of past worker reallocation to future generations through the relative wage.

Lemma 3. *Past worker reallocation induces an indirect worker-allocation response through the period-$t+1$ relative wage when the economy features intergenerational persistence.*

$$\mathcal{F}^{indirect,IGP\text{-}\kappa} \quad > \quad \mathcal{F}^{indirect,IGP\text{-}\tau} \quad > \quad \mathcal{F}^{indirect,bench} \quad = \quad 0$$

The magnitude of this transmission is larger when the economy features inherited productivity than when it features inherited immobility.

With fewer ag-born workers due to a previous allocation of workers out of agriculture, the effective labor supply tilts toward nonagriculture, which leads to a reduction in the

relative wage. For the IGP-τ economy, the relative-wage reduction offsets part, but not all, of the direct propagation of past worker reallocation. Thus workers leaving agriculture in the current generation induce workers to leave agriculture in the next generation. In the IGP-κ economy, the relative-wage reduction is stronger and may or may not overcome the direct propagation channel. Consequently, the sign of the total response is ambiguous and depends on the relative magnitudes of the two channels.

Proposition 3. *Past worker reallocation alters the composition of ag-born and nonag-born workers in future generations, which in turn affects the aggregate worker allocation in those generations.*

$$\left(\frac{d\phi_{A,t+1}}{d\phi_{A,t}}\right)^{\text{IGP-}\tau} > 0, \qquad \left(\frac{d\phi_{A,t+1}}{d\phi_{A,t}}\right)^{\text{bench}} = 0.$$

In the IGP-κ economy, the derivative can be positive or negative depending on whether the direct composition effect or the induced relative-wage effect dominates.

I illustrate the ambiguity of the dynamic transmission of past worker reallocation in the IGP-κ economy in figure 7 below. For most empirically relevant values of the inherited productivity wedge, the direct composition effect is dominated by the induced relative-wage effect, and past worker reallocation out of agriculture induces future generations to return to agriculture. For large enough κ values, the direct composition effect dominates, and past worker reallocation out of agriculture induces future generations to leave agriculture as well. In contrast, the IGP-τ economy always features a positive dynamic derivative, so past worker reallocation out of agriculture induces future generations to leave agriculture as well.

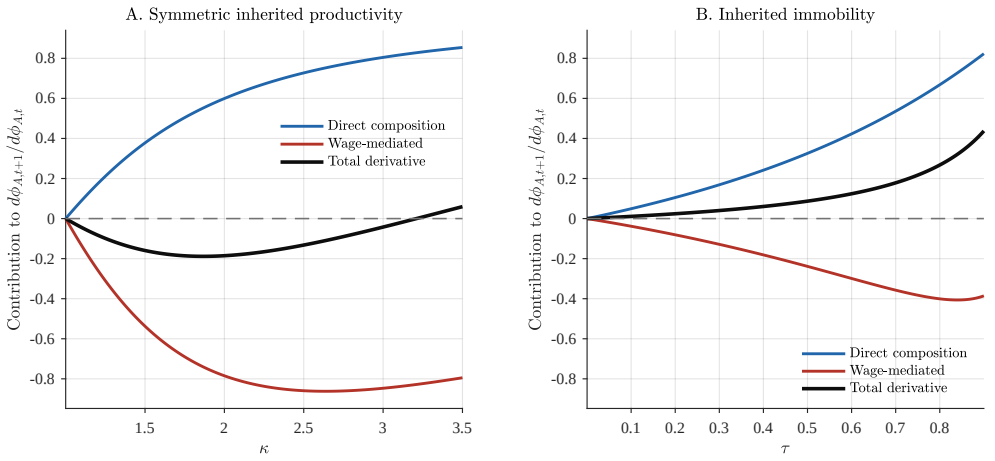


Figure 7: How Does Past Worker Reallocation Affect Future Generations?

Notes: The figure plots $d\phi_{A,t+1}/d\phi_{A,t}$ under homothetic demand, with $\phi_A = 0.665$, $\gamma = 2.076$, and $\sigma = 0.3$. Panel A varies symmetric inherited productivity, $\kappa_A = \kappa_N = \kappa \in (1, 3.5]$, with $\tau = 0$; Panel B varies inherited immobility, $\tau \in (0, 0.9]$, with $\kappa_A = \kappa_N = 1$. At each value, I solve for the stationary allocation gap, which implies an agricultural-employment odds ratio of $\kappa^{2\gamma}$ in Panel A and $(1-\tau)^{-\gamma}$ in Panel B. Blue and red show the direct composition and wage-mediated effects, respectively; black shows their sum, and the dashed line marks zero. Positive values imply that the next generation continues the initial reallocation out of agriculture; negative values imply partial reversal.

Discussion Intergenerational persistence alters the dynamics of structural transformation playing out over multiple new generations. For reallocation out of agriculture driven by a permanent improvement in agricultural technology, the newest generation arriving to the workforce at the same time as the shock reallocates less with intergenerational persistence than without it. In particular, inherited immobility reduces this immediate departure from agriculture even more than inherited productivity. However, intergenerational persistence also propagates the effects of past reallocation forward to future generations. As a result, with inherited immobility, the first generation's lost reallocation is partially recovered in the next generation. But with inherited productivity, the next generation is induced to return to agriculture, which compounds on the immediate effect.

The common stationary allocation, matched allocation gap, pure persistence regimes, symmetric productivity wedges, and homothetic demand allow the analytical exercise to isolate the core channels through which intergenerational persistence shapes structural transformation. The quantitative application that follows in the next section relaxes these restrictions. In particular, I allow inherited productivity and inherited immobility to operate jointly, permit $\kappa_A \neq \kappa_N$, and restore nonhomothetic demand.

4. Quantitative Application: U.S. Industrialization

Here I present the quantitative application to the historical episode of U.S. structural transformation out of agriculture between 1880 and 1940. First, I calibrate the production-side parameters of the model to obtain the closest match to the historical evolution of agricultural output share, relative price, and real GDP per capita. Then, I identify the sectoral technologies and the relative payoff shifter that exactly replicate the historical evolution of agricultural employment share, relative price, and real GDP per capita. Finally, I simulate the model under counterfactual regimes of intergenerational persistence to quantify how inherited productivity and inherited immobility influence the historical reallocation of workers out of agriculture.

4.1. Production-Side Parameter Calibration

I externally set $\sigma = 0.3$, consistent with the estimates in Comin, Lashkari and Mestieri (2021). I normalize $\Omega_N = 1$ and $\vartheta_A = 1$, calibrate Ω_A from the initial agricultural output share, and choose $\vartheta_N > \vartheta_A$ by minimum distance. I observe the agricultural output share, $\omega_{A,t}$, from Dennis and İşcan (2009), the relative price $p_t \equiv p_{N,t}/p_{A,t}$ from Alvarez-Cuadrado and Poschke (2011), and real GDP per capita, Y_t/P_t, from Maddison (1995) for each year $t \in \{1880, 1900, 1920, 1940\}$. I scale both the relative price and real GDP per capita to be equal to one in 1880, so that the 1880 agricultural output share determines the relative demand shifters.

By normalizing $\Omega_N = 1$, I calibrate Ω_A to match the 1880 agricultural output share:

$$\Omega_A = \frac{\omega_{A,1880}}{\omega_{N,1880}}$$

Then, I choose ϑ_N by minimum distance to the observed agricultural output shares:

$$\widehat{\vartheta}_N = \arg\min_{\vartheta_N > 1} \sum_{t \in \{1900,1920,1940\}} \left(\omega_{A,t} - \widehat{\omega}_{A,t}(\vartheta_N; \sigma = 0.3)\right)^2.$$

The model-implied relative output share is

$$\frac{\widehat{\omega}_{A,t}}{1 - \widehat{\omega}_{A,t}} = \Omega_A \left(\frac{p_{N,t}}{p_{A,t}}\right)^{\sigma-1} \left(\frac{Y_t}{P_t}\right)^{(1-\vartheta_N)(1-\sigma)}.$$

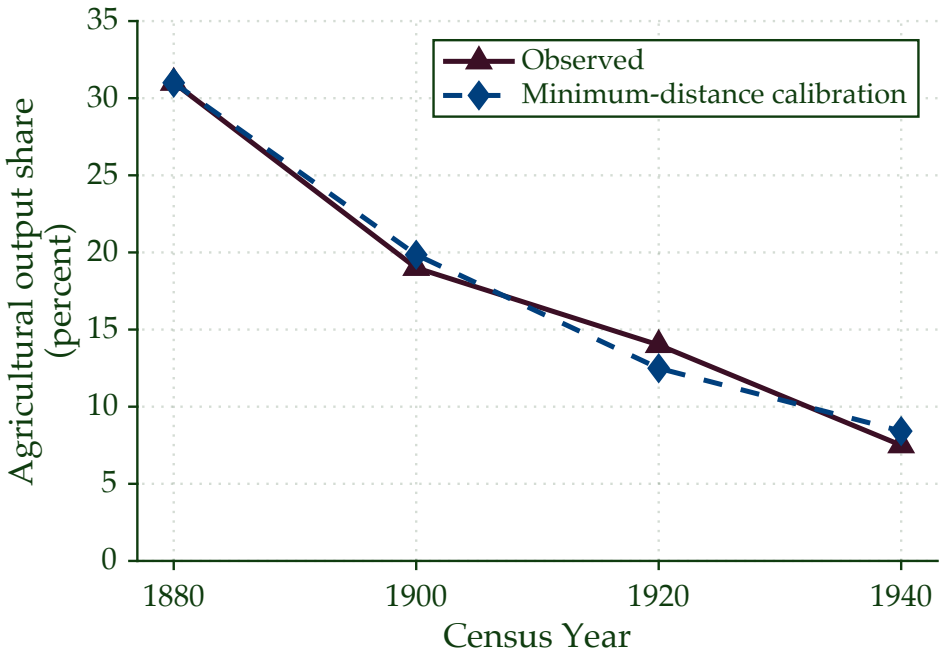


Figure 8: Production-Side Calibration of the Agricultural Output Share

Note: The figure compares the observed agricultural output share with the baseline minimum-distance calibration. The baseline fixes $\sigma = 0.3$ and $\vartheta_A = 1$, estimates $\vartheta_N = 4.630$ from the 1900–1940 shares, normalizes $\Omega_N = 1$, and chooses Ω_A to match the 1880 share. Output shares, relative prices, and real GDP per capita are from Dennis and İşcan (2009), Alvarez-Cuadrado and Poschke (2011), and Maddison (1995), respectively.

The minimum-distance estimate of the nonagricultural nonhomotheticity parameter is $\widehat{\vartheta}_N = 4.63$. Figure 8 plots the observed and model-implied agricultural output shares from the baseline calibration. By design, the model exactly matches the observed agricultural output share in 1880, and its largest discrepancy thereafter is 1.5 percentage points in 1920. Appendix C.1.1 reports the fit and counterfactual results under an external production-side calibration, alternative labor-supply estimates, and wedges estimated under alternative skill distributions.

Table 4 summarizes the benchmark parameters used in the quantitative analysis and the empirical discipline for each value.

Table 4: Benchmark Model Parameters

Parameter	Value	Empirical discipline
Panel A: Labor-supply parameters		
γ	2.076	Dispersion of log hourly earnings in the 1940 linked-census sample within worker-parent sector cells.
κ_A	1.604	Selection-adjusted hourly-earnings gap between ag-born and nonag-born workers employed in agriculture.
κ_N	1.797	Selection-adjusted hourly-earnings gap between nonag-born and ag-born workers employed in nonagriculture.
τ	0.328	Residual intergenerational sector-choice odds gap after accounting for κ_A and κ_N.
Panel B: Production-side parameters		
Ω_A	0.449	Chosen to match the 1880 agricultural output share.
Ω_N	1	Normalization.
σ	0.300	Set externally, consistent with the estimates in Comin, Lashkari and Mestieri (2021).
ϑ_A	1	Normalization.
ϑ_N	4.630	Minimum-distance fit to the agricultural output shares in 1900, 1920, and 1940.
Panel C: Initial Conditions		
$\phi_{A,0}$	0.665	Share of ag-born workers in 1880.

Note: Panel A reports the baseline labor-supply parameters: γ and the earnings moments in κ_A and κ_N use 1940 linked-census workers with valid hourly earnings and at least 35 weekly hours, without restricting log earnings; the sector-choice moments in κ_A, κ_N, and τ use the full linked sample. Panel B reports the production-side calibration, and Panel C the initial share of ag-born workers entering in 1880, equal to the 1860 agricultural employment share.

4.2. Inferring Drivers of U.S. Structural Transformation

I exploit the model's structure along with observed agricultural employment shares, relative prices, and real GDP per capita to infer the sectoral technologies and relative payoff shifter that exactly replicate the historical evolution of U.S. structural transformation out of agriculture.

The inversion is period-by-period but recursive because the agricultural employment share in period $t-1$ determines the fraction of workers in period t who are ag-born. Conditional on this inherited composition and the calibrated parameters, the three observed

outcomes in period t identify the three fundamentals $Z_{A,t}$, $Z_{N,t}$, and $B_{N,t}/B_{A,t}$.

Proposition 4 (Identification by model inversion)**.** *Fix the following model parameters.*

$$\{\gamma, \kappa_A, \kappa_N, \tau, \sigma, \vartheta_A, \vartheta_N, \Omega_A, \Omega_N\}.$$

Assume $\gamma > 1$, $\tau \in [0, 1)$, $\sigma \in (0, 1)$, and strictly positive productivity and demand weights. Normalize $p_{A,t} = 1$ and take the inherited worker composition $\phi_{A,t-1}$ as known. Consider the following interior observed outcomes.

$$\left\{\phi_{A,t},\ p_t \equiv \frac{p_{N,t}}{p_{A,t}},\ Q_t \equiv \frac{Y_t}{P_t}\right\} \in (0, 1) \times \mathbb{R}^2_{++}$$

Any observed outcomes in this set that are consistent with the model uniquely identify the period-t fundamentals $\{Z_{A,t}, Z_{N,t}, B_{N,t}/B_{A,t}\} \in \mathbb{R}^3_{++}$.

The result follows from a sequence of closed-form mappings after solving one strictly monotone scalar equation. First, $\phi_{A,t}$ uniquely determines the product of $B_{N,t}/B_{A,t}$ and $w_{N,t}/w_{A,t}$, and hence the sector choices and effective labor supplies of ag-born and nonag-born workers. Second, relative goods demand determines $Z_{N,t}/Z_{A,t}$ from p_t, Q_t, and those effective labor supplies. Third, the level of real GDP pins down the levels of sectoral technologies through the final-good aggregator. Finally, $w_{N,t}/w_{A,t} = p_t Z_{N,t}/Z_{A,t}$ separates the relative payoff shifter from the relative wage. Appendix C.1 provides the complete inversion and proof.

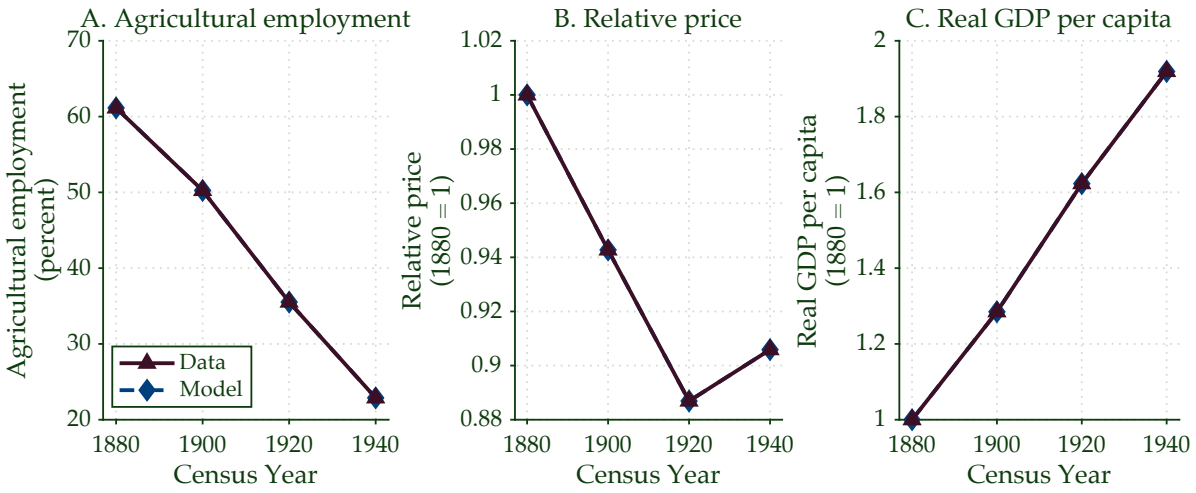


Figure 9: Targeted Moments for Model Inversion

Note: The figure compares the data with the outcomes reproduced by the period-by-period model inversion. Agricultural employment is reported in percent; the relative price, p_N/p_A, and real GDP per capita are indexed to one in 1880. The inversion targets all three series exactly in 1880, 1900, 1920, and 1940, recursively using the previous period's agricultural employment share as the current share of ag-born workers. Employment shares are calculated from the linked censuses; relative prices and real GDP per capita are from Alvarez-Cuadrado and Poschke (2011) and Maddison (1995), respectively.

In Figure 9, I plot the observed agricultural employment share, relative price, and real GDP per capita that are used to identify the sectoral technologies and relative payoff shifter in each year. While the inversion exactly matches the aggregate agricultural employment share, it generally does not match the subgroup employment shares of ag-born and nonag-born workers. I plot observed and model-implied subgroup agricultural employment shares in Figure 10. This mismatch emerges because the extent of intergenerational persistence changes somewhat over time, but my calibration holds the inherited productivity and inherited immobility wedges constant at their 1940 values. Most notably, the model implies a much higher agricultural employment share for nonag-born workers in 1880 than is observed in the data.

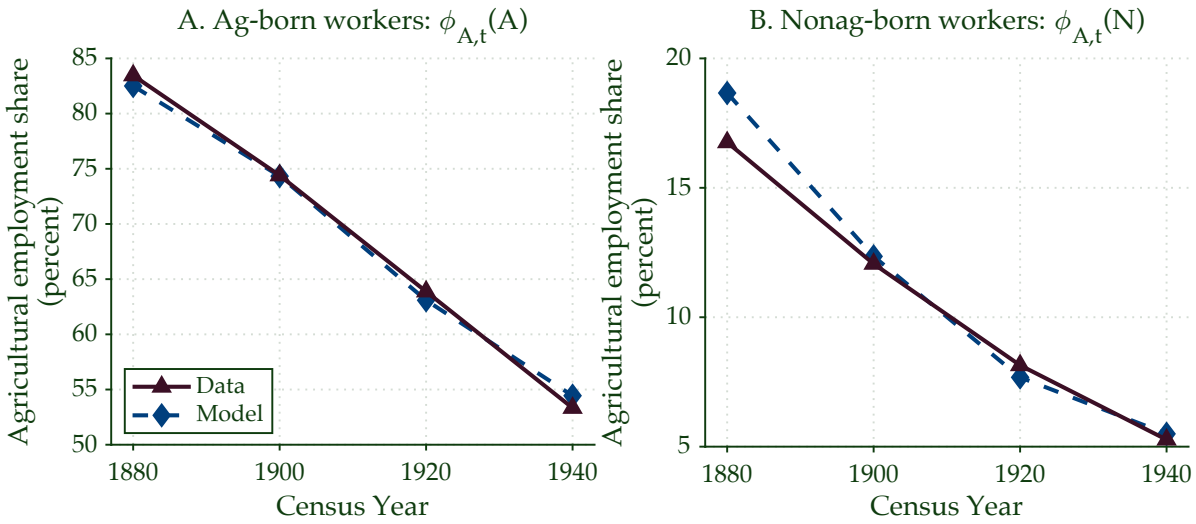


Figure 10: Agricultural Employment Share for Ag-born and Nonag-born Workers

Note: The figure compares observed and model-implied agricultural employment shares for ag-born and nonag-born workers in the linked censuses. These subgroup shares are not targeted: the inversion matches only the aggregate employment share, relative price, and real GDP per capita in each year, using the time-invariant baseline labor-supply parameters estimated from 1940.

Within the current model, matching the subgroup paths would require time-varying intergenerational-persistence wedges or an additional mechanism. Without income microdata prior to 1940, estimating such time variation would require maintained assumptions about the evolution of intergenerational persistence that are not supported by data. I therefore interpret the inferred fundamentals as those required to rationalize the aggregate transition conditional on holding the 1940 intergenerational persistence wedges fixed, rather than as a decomposition that separately identifies historical changes in persistence.

The inferred sectoral technologies and relative payoff shifter, each indexed to their 1880 values, are plotted in Figure 11. I infer dramatic growth in the nonagricultural and, to a lesser extent, the agricultural technology between 1880 and 1940. In particular, the nonagricultural technology is 11 times larger in 1940 than in 1880, while the agricultural technology is about 6.6 times larger. The relative payoff shifter also contributes to the reallocation of workers out of agriculture, but its evolution is much more modest: about 30 percent larger in 1940 than in 1880.

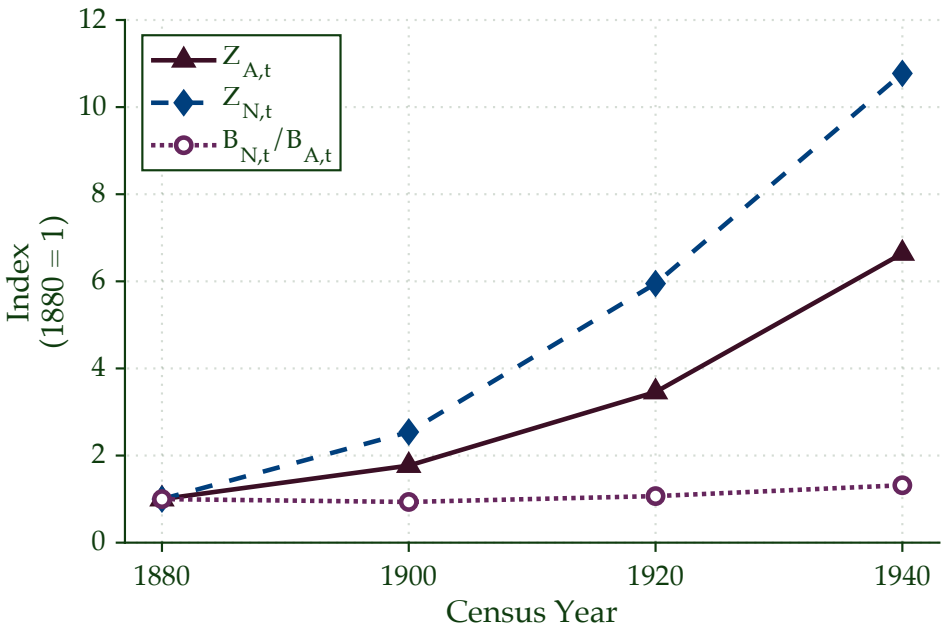


Figure 11: Inferred Drivers of U.S. Structural Transformation, 1880-1940

Note: The figure plots the agricultural and nonagricultural technologies, $Z_{A,t}$ and $Z_{N,t}$, and the relative payoff shifter, $B_{N,t}/B_{A,t}$, inferred from the targeted moments in Figure 9. Each series is indexed to one in 1880; subsequent values are recovered recursively under the baseline parameters in Table 4.

Because these fundamentals are inferred through the model, their measured growth depends on whether the inversion accounts for intergenerational persistence. Figure 12 repeats the inversion after imposing no inherited productivity or inherited immobility, while holding γ and the production-side parameters fixed. Without IGP, matching the same observed outcomes requires substantially faster relative nonagricultural technology growth: the 1880–1940 increase in $Z_{N,t}/Z_{A,t}$ rises from 48.4 to 82.1 log points. Conversely, the inferred increase in $B_{N,t}/B_{A,t}$ falls from 27.6 to 8.1 log points. Thus an inversion that omits IGP attributes more of the observed worker reallocation to relative technology growth and less to changes in relative sectoral payoffs. For comparison, the nonfarm-to-farm relative TFP series constructed by Alvarez-Cuadrado and Poschke (2011) rises by 55.5 log points over the same period, much closer to the baseline than the no-IGP inversion.

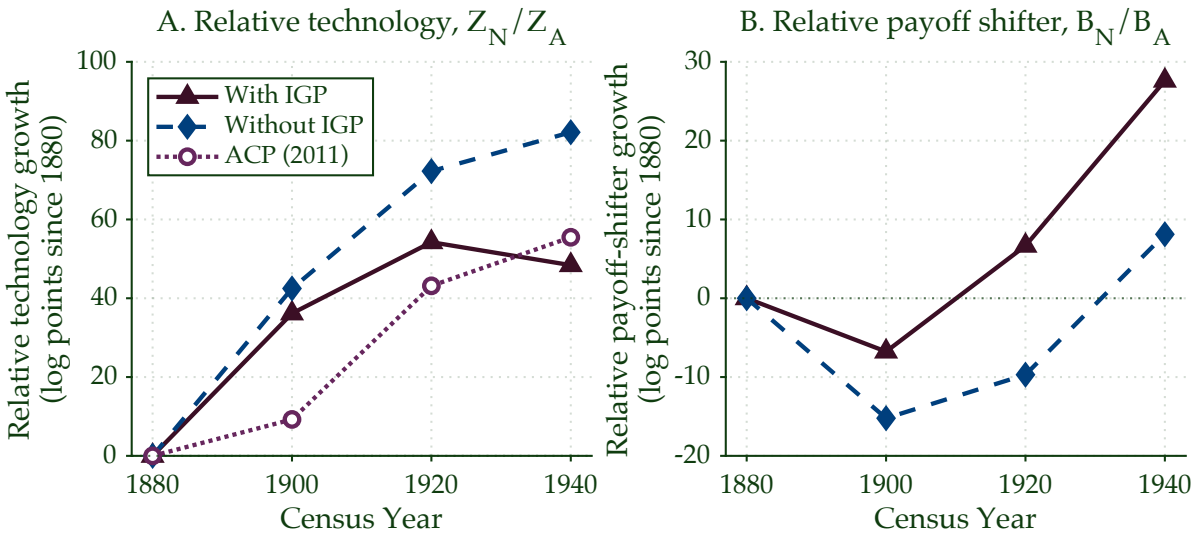


Figure 12: Inferred Relative Drivers With and Without Intergenerational Persistence

Note: Panel A plots the cumulative log change in relative technology, $Z_{N,t}/Z_{A,t}$; Panel B plots the cumulative log change in the relative payoff shifter, $B_{N,t}/B_{A,t}$. Both model series are normalized to zero in 1880. Panel A also plots the U.S. nonfarm-to-farm relative TFP series from the replication data of Alvarez-Cuadrado and Poschke (2011). Because that series is observed in 1879 and 1884 but not 1880, I linearly interpolate its 1880 level; its values in 1900, 1920, and 1940 are observed directly. The baseline inversion uses the estimated inherited-productivity and inherited-immobility wedges. The no-IGP inversion imposes $\kappa_A = \kappa_N = 1$ and $\tau = 0$, while holding γ and all production-side parameters fixed. Both inversions match the same agricultural employment share, relative price, and real GDP per capita in each year.

Appendix Figure 15 decomposes this difference across the two sources of persistence. Removing inherited productivity alone raises inferred relative nonagricultural technology growth from 48.4 to 80.7 log points, nearly the 82.1 log points inferred without any intergenerational persistence, whereas removing inherited immobility raises it only to 51.6 log points. Thus, the change in the model's interpretation of U.S. industrialization comes primarily from accounting for inherited productivity rather than inherited immobility.

4.3. U.S. Structural Transformation in Counterfactual IGP Regimes

In order to quantify the role of inherited productivity and inherited immobility in shaping U.S. structural transformation, I simulate the model under counterfactual regimes of intergenerational persistence. I consider three counterfactual regimes. First, I remove inherited productivity by setting $\kappa_A = \kappa_N = 1$ while retaining the inherited immobility of ag-born workers by keeping $\tau = 0.328$. Second, I remove inherited immobility by setting $\tau = 0$ while retaining inherited productivity at $\kappa_A = 1.604$ and $\kappa_N = 1.797$. Third, I remove both sources of intergenerational persistence by setting $\kappa_A = \kappa_N = 1$ and $\tau = 0$.

For each counterfactual regime, I re-infer the sectoral technologies and relative payoff shifter that exactly replicate the observed agricultural employment share, relative price, and real GDP per capita in 1880. Then, I simulate the model forward under the counterfactual regime of intergenerational persistence using the inferred growth in sectoral technologies and relative payoff shifter depicted in Figure 11. By construction, the counterfactual simulations exactly match the observed outcomes in 1880, but they generally do not match the observed outcomes thereafter.

Table 5: U.S. Worker Reallocation under Counterfactual IGP Regimes, 1880–1940

	Worker reallocation	Difference from benchmark	
IGP regime	Percentage points	Percentage points	Percent
Benchmark	38.3	–	–
No inherited productivity	44.2	5.9	15.5
No inherited immobility	39.1	0.8	2.1
No intergenerational persistence	43.7	5.5	14.3

Note: Worker reallocation is the 1880–1940 decline in the agricultural employment share; differences are relative to the benchmark, in percentage points and as a percent of benchmark reallocation. Each counterfactual is re-anchored to the 1880 employment share, relative price, and real GDP per capita, then simulated using benchmark growth in the inferred technologies and relative payoff shifter.

Table 5 compares the decline in the agricultural employment share between 1880 and 1940 under each counterfactual regime to the benchmark/observed decline. Removing inherited productivity increases worker reallocation out of agriculture by 5.9 percentage points, or 15.5 percent of benchmark reallocation. Removing inherited immobility increases worker reallocation by 0.8 percentage points, or 2.1 percent of benchmark reallocation. Removing both sources of intergenerational persistence increases worker reallocation by 5.5 percentage points, or 14.3 percent of benchmark reallocation.[19]

I next decompose the effect of each counterfactual IGP regime on worker reallocation into the two channels highlighted by the analytical results: workers' contemporaneous response to changing fundamentals and the intergenerational transmission of past reallocation through parental composition. These two channels interact in equilibrium; hence, their measured contributions depend on the order in which they are changed. As such, I

[19]For reference, the U.S. industrialization lost to intergenerational persistence is comparable to the Chinese industrialization spurred by modern labor mobility reforms. Brandt, Hsieh and Zhu (2008) estimate that the 1978–2004 decline in China's agricultural employment share would have been 8 percentage points (21 percent) smaller without the observed reductions in agricultural versus nonagricultural wage gaps that they attribute to mobility reforms.

use a Shapley decomposition, which assigns each channel its average marginal contribution across the possible orderings. I implement this decomposition within each historical interval and sum the resulting contributions over 1880–1940.

Table 6: Transmission Channels under Counterfactual IGP Regimes

IGP regime	Contemporaneous response	Intergenerational propagation	Total
	Percentage points		
No inherited productivity	2.63	3.31	5.94
No inherited immobility	3.23	−2.43	0.80
No intergenerational persistence	3.71	1.76	5.47

Note: The interval-level Shapley decomposition averages over the two orders in which fundamentals and parental composition change. The contemporaneous channel varies technologies and the relative payoff shifter holding composition fixed; the intergenerational channel varies composition holding those fundamentals fixed. Entries are cumulative 1880–1940 contributions relative to full IGP and sum to the counterfactual reallocation differences in Table 5.

Table 6 shows that removing inherited productivity increases worker reallocation through both channels, with intergenerational propagation accounting for 3.31 of the total 5.94 percentage point effect. Removing inherited immobility generates a large contemporaneous increase in worker reallocation that is mostly offset by the resulting change in parental composition. When I remove both sources of intergenerational persistence, the contemporaneous response increases worker reallocation by 3.71 percentage points, while intergenerational propagation contributes 1.76 percentage points. Appendix Table 16 shows that these channel results are qualitatively stable across alternative production-side and labor-supply calibrations, though the magnitudes vary. Appendix Table 18 further shows that the mechanism-specific pattern persists when I replace the inherited wedges with those estimated under bivariate lognormal and Fréchet-Frank skill distributions, while retaining the baseline Fréchet quantitative structure. Under both alternatives, removing inherited productivity produces a larger cumulative increase in reallocation than removing inherited immobility, whose larger contemporaneous effect is substantially offset by intergenerational propagation.

5. Conclusion

Though new generations provide crucial flexibility for labor market adjustment, their aggregate responsiveness need not equal the primitive responsiveness of the ag-born and

nonag-born groups that comprise them. In particular, when workers inherit productive advantages in their parents' sector, then they are much less responsive to wage changes, which reduces the newest generations' departure from agriculture. This drag on transformation is moreover compounded by the next generation who are induced to return to agriculture. On the other hand, inherited immobility is a greater immediate impediment to worker reallocation, but that is partially offset by the next generation induced to leave agriculture. Thus, parental sector dependence changes both the economy's response to structural change forces and the forces a quantitative model infers from the observed transition.

One direction for future work is to ask how intergenerational persistence shapes structural transformation in developing economies today. Workers in these economies face substantial barriers to moving across sectors and locations (Bryan and Morten, 2019; Donovan and Schoellman, 2023), and agricultural occupations are often strongly correlated across generations (Emran and Shilpi, 2011). Moreover, despite substantial agricultural productivity growth following the Green Revolution, labor reallocation out of agriculture has lagged (Krishna and Meenakshi, 2022). My findings suggest that intergenerational persistence may be an important factor in this delay, especially if inherited productivity is a significant source of persistence. In the context of India, Cassan, Keniston and Kleineberg (2021) findings are consistent with even greater inherited productivity than in my setting of the historical U.S.

A second direction is to incorporate intergenerational persistence into models of adjustment to shorter-run sectoral shocks. For example, the China trade shock caused large and persistent employment losses in exposed U.S. labor markets (Autor, Dorn and Hanson, 2013). Several papers have developed quantitative structural models to study these adjustments usually featuring Roy frameworks with Fréchet distributed skills or preferences (Caliendo, Dvorkin and Parro, 2019; Galle, Rodríguez-Clare and Yi, 2023). A core insight from my analysis is that workers' aggregate responsiveness to wage changes is not fully encapsulated by the Fréchet shape parameter when the economy features intergenerational persistence. Accounting for new workers' inherited connections to their parents' sector may therefore be important for calibrating these models and also for understanding the transitions the study. Indeed, Autor et al. (2025) present evidence that the eventual recovery of exposed places came disproportionately from workers who were below working age when the shock began, rather than from incumbent workers changing sectors. Embedding inherited productivity and immobility in a dynamic trade model could therefore reveal whether adjustment by new cohorts attenuates a sectoral shock or

carries its effects into the next generation.

Intergenerational mobility has become a central concern for researchers and policymakers seeking to understand the persistence of inequality and the scope for policy to expand economic opportunity (OECD, 2018; Narayan et al., 2018).[20] This paper emphasizes a different consequence of persistence: its effect on an economy's capacity to reallocate workers over long horizons. The results show that this effect can be substantial and depends critically on whether persistence reflects inherited productivity or inherited immobility. As artificial intelligence and other technologies reshape the demand for workers across occupations and sectors (Acemoglu and Restrepo, 2019) in modern economies, forecasts that treat new workers as independent of the previous generation may misstate both the speed and the persistence of labor-market adjustment. Researchers and policymakers anticipating these changes should therefore consider not only whether incumbent workers can move, but also how the skills and attachments inherited by new workers shape where they enter the economy.

[20] Much of this literature measures mobility using income or educational attainment rather than occupation or sector of work; prominent examples include Black and Devereux (2011), Chetty et al. (2014), Chetty and Hendren (2018*a*), Chetty and Hendren (2018*b*), and Chetty et al. (2020). Beyond the studies cited earlier, recent work on the determinants of intergenerational occupational mobility includes Bell et al. (2019) and Greenberg et al. (2024).

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A. Appendix: Empirical Results

A.1. Data Construction

This appendix assesses the robustness of the linked-census estimates to the procedure used to link sons to their childhood households.

A.1.1. MLP Sample Selection

Appendix Table 7 documents how the MLP linkage procedure changes the analysis sample. Panel A begins with U.S.-born employed men ages 20–29 with a valid current industry and reports the number linked to a childhood census record, the number for whom a father is identified in the childhood household, and the number with a valid paternal industry. The final sample ranges from 49 percent of eligible workers in 1880 to 68 percent in 1940. Panel B compares the full eligible population with the final linked sample in 1940. Despite the reduction in sample size, the two samples are similar along the observed dimensions reported in the table. The linked sample is modestly more agricultural, white, and educated and modestly less urban and Southern, while its sector-specific earnings moments remain close to those in the full sample.

Table 7: MLP Linkage and Sample Selection

Panel A: MLP sample construction

Year	Eligible	Childhood link	Father identified	Final sample	Final share
1880	1,756,491	1,152,891	1,047,863	864,166	49.2%
1900	2,619,056	1,644,774	1,565,404	1,394,395	53.2%
1920	4,786,523	3,638,704	3,343,024	2,971,796	62.1%
1940	7,393,949	5,903,317	5,491,663	5,022,389	67.9%

Panel B: Full versus linked sample, 1940

Statistic	Full eligible sample	Final MLP sample	Difference
Agricultural employment share	0.215	0.229	0.014
Mean age	24.556	24.264	-0.291
White share	0.923	0.941	0.018
Urban share	0.536	0.518	-0.017
High-school graduate share	0.381	0.408	0.027
South share	0.290	0.273	-0.016
Valid hourly-earnings share	0.634	0.621	-0.013
Mean log hourly earnings: Agriculture	-1.931	-1.944	-0.013
Variance of log hourly earnings: Agriculture	0.899	0.913	0.015
Mean log hourly earnings: Nonagriculture	-0.778	-0.783	-0.005
Variance of log hourly earnings: Nonagriculture	0.434	0.429	-0.005

Note: Panel A begins with U.S.-born employed men ages 20–29 with a valid current industry. "Childhood link" requires an MLP link to the relevant preceding census, using 1880 as the childhood census for workers observed in 1900 because the 1890 census was destroyed. "Father identified" additionally requires a household-head father in the childhood record, and the final sample additionally requires a valid paternal industry. The final share is relative to the eligible population in the first column. Panel B compares the full eligible population with the final MLP sample in 1940, and the final column reports the linked-sample mean minus the full-sample mean. The hourly-earnings sample requires positive valid wage income, weeks worked, and usual weekly hours; recorded topcoded income of $5,001 issettoits$5,000 censoring threshold.

A.1.2. Alternative Historical Linkages

My baseline analysis uses the Multigenerational Longitudinal Panel (MLP) links distributed by IPUMS. As an alternative, I use the Census Tree crosswalks to link each adult observation to his childhood census record and then identify his father as the household head in that census. I also repeat the analysis using the automated linking procedure of Abramitzky et al. (2021), hereafter ABE. My primary ABE results use exact-name conservative links, which require a unique match within a two-year age window, and I additionally report the less restrictive exact-name standard links. I apply the same age, sex, birthplace, employment, industry, earnings, schooling, and geographic restrictions under all linkage procedures.

Appendix Table 8 compares the resulting estimates. The alternative links produce somewhat smaller sector-choice gaps, particularly in 1880 and 1900, but the differences narrow substantially by 1920 and 1940. The primary ABE estimates are especially close

to the Census Tree estimates, while the larger ABE standard sample produces somewhat smaller gaps. The earnings gaps, Fréchet shape parameter, and inherited-productivity wedges are also similar across linkage procedures. The inferred immobility wedge is modestly smaller using Census Tree and ABE, but the substantive conclusion that both inherited productivity and inherited immobility contribute to sectoral persistence is unchanged.

Table 8: Robustness to Alternative Historical Linkages

Outcome	Specification	MLP	Census Tree	ABE conservative	ABE standard
Panel A: Sector choice					
Sector choice, 1880	Unconditional	0.667	0.594	0.583	0.537
Sector choice, 1880	Full controls	0.556	0.466	0.468	0.412
Sector choice, 1900	Unconditional	0.621	0.563	0.545	0.511
Sector choice, 1900	Full controls	0.499	0.421	0.422	0.375
Sector choice, 1920	Unconditional	0.558	0.532	0.521	0.488
Sector choice, 1920	Full controls	0.445	0.416	0.417	0.379
Sector choice, 1940	Unconditional	0.482	0.457	0.445	0.424
Sector choice, 1940	Full controls	0.396	0.373	0.367	0.345
Panel B: Log hourly earnings					
Log hourly earnings, Agricultural workers	Unconditional	-0.677	-0.616	-0.600	-0.530
Log hourly earnings, Nonagricultural workers	Unconditional	-0.322	-0.316	-0.297	-0.297
Log hourly earnings, Agricultural workers	Full controls	-0.436	-0.395	-0.399	-0.350
Log hourly earnings, Nonagricultural workers	Full controls	-0.173	-0.166	-0.165	-0.158
Panel C: Structural parameters					
$\widehat{\gamma}$	Baseline	2.076	2.053	2.082	2.071
$\widehat{\kappa}_A$	Baseline	1.604	1.632	1.597	1.585
$\widehat{\kappa}_N$	Baseline	1.797	1.759	1.685	1.667
$\widehat{\tau}$	Baseline	0.328	0.295	0.308	0.269
$\widehat{\gamma}$	Added fixed effects	2.293	2.270	2.302	2.293
$\widehat{\kappa}_A$	Added fixed effects	1.322	1.340	1.324	1.319
$\widehat{\kappa}_N$	Added fixed effects	1.433	1.408	1.385	1.363
$\widehat{\tau}$	Added fixed effects	0.284	0.247	0.257	0.224

Note: The table compares estimates obtained from the baseline IPUMS MLP links with estimates obtained from Census Tree and ABE links. ABE conservative uses exact-name links that are unique within a two-year age window, while ABE standard uses the less restrictive exact-name procedure. Panel A reports linear-probability coefficients on an indicator for a father employed in agriculture, where the outcome is an indicator for the son working in agriculture. Full controls in Panel A absorb childhood schooling, childhood commuting zone, and adulthood commuting zone. Panel B reports coefficients on father employed in agriculture from regressions of log hourly earnings estimated separately for agricultural and nonagricultural workers in 1940. Full controls in Panel B absorb childhood schooling, childhood commuting zone, and adulthood commuting zone. Panel C reports the Fréchet shape parameter and inferred wedges using harmonized samples and definitions. Its added-fixed-effect specification includes adulthood commuting-zone and age fixed effects, matching the corresponding specification in the main analysis.

A.2. Labor Supply Parameter Estimates

This appendix provides additional details and robustness checks for the labor-supply parameter estimates in Section 2. I first establish the conditional variance result used to identify γ and decompose log-odds persistence between the two wedges. I then assess alternative treatments of topcoded earnings and conclude by considering annual and weekly earnings.

A.2.1. Conditional Log-Income Variance

Fix a parental sector k, and write indirect utility in sector j as $V_{i,j,t}(k) = a_{j,t}(k)\epsilon_{i,j,t}$, where $a_{j,t}(k)$ collects the wage, payoff shifter, and applicable inherited wedges. Because $\epsilon_{i,j,t}^{-\gamma}$ is standard exponential,

$$T_{i,j,t}(k) \equiv V_{i,j,t}(k)^{-\gamma} \sim \text{Exp}(a_{j,t}(k)^{\gamma})$$

independently across sectors. Maximizing $V_{i,j,t}(k)$ is therefore equivalent to winning an exponential race by minimizing $T_{i,j,t}(k)$. For independent exponentials, the minimum $T_{i,t}^{\min}(k)$ is independent of the identity of the minimizing sector and has rate $\sum_j a_{j,t}(k)^{\gamma} = \Lambda_t(k)^{\gamma}$. Thus $V_{i,t}^{\max}(k) = [T_{i,t}^{\min}(k)]^{-1/\gamma}$ is independent of sector choice, and, because log income conditional on (j,k) differs from $\log V_{i,t}^{\max}(k)$ only by a constant,

$$\text{Var}[\log y_{i,j,t}(k) \mid s_{i,t} = j,\ s_{i,t-1} = k] = \frac{\pi^2}{6\gamma^2}.$$

A.2.2. Decomposing Log-Odds Persistence

Equation (ID-4) decomposes the log intergenerational odds ratio additively as

$$\log \mathcal{O} = \underbrace{\gamma\left(\log \kappa_A + \log \kappa_N\right)}_{\text{inherited productivity}} + \underbrace{-\gamma \log(1-\tau)}_{\text{inherited immobility}}.$$

I define each wedge's contribution to persistence as its component divided by $\log \mathcal{O}$. Using $(\widehat{\gamma}, \widehat{\kappa}_A, \widehat{\kappa}_N, \widehat{\tau}) = (2.076, 1.604, 1.797, 0.328)$ and $\widehat{\mathcal{O}} = 20.541$ gives

$$\widehat{s}_{\kappa} = \frac{\widehat{\gamma}(\log \widehat{\kappa}_A + \log \widehat{\kappa}_N)}{\log \widehat{\mathcal{O}}} = 0.727, \qquad \widehat{s}_{\tau} = \frac{-\widehat{\gamma}\log(1-\widehat{\tau})}{\log \widehat{\mathcal{O}}} = 0.273.$$

Thus inherited productivity accounts for 72.7 percent of observed log-odds persistence and inherited immobility accounts for the remaining 27.3 percent.

A.2.3. Inherited Wedges among Movers

The main estimates may partly reflect systematic differences in the local labor markets occupied by ag-born and nonag-born workers or advantages conferred by local family networks. To assess this concern, I repeat the added-fixed-effect procedure among workers whose adult commuting zone differs from their childhood commuting zone. The Movers column of Table 9 uses mover-specific hourly-earnings moments and sector-choice probabilities, while retaining the added-fixed-effect estimate of the Fréchet shape parameter. The inherited productivity and immobility wedges become smaller, but remain economically meaningful among movers.

Table 9: Inherited Productivity and Immobility Wedges among Movers

	Baseline	Added FE	Movers
Panel A: Inferred wedges			
Inherited ag. productivity, $\widehat{\kappa}_A$	1.604	1.323	1.188
Inherited nonag. productivity, $\widehat{\kappa}_N$	1.797	1.437	1.283
Inherited mobility wedge, $\widehat{\tau}$	0.328	0.281	0.141
Panel B: Mean hourly earnings			
Agriculture, ag-born: $\widehat{y}_A(A)$	0.19	0.20	0.22
Agriculture, nonag-born: $\widehat{y}_A(N)$	0.37	0.32	0.29
Nonagriculture, ag-born: $\widehat{y}_N(A)$	0.47	0.52	0.53
Nonagriculture, nonag-born: $\widehat{y}_N(N)$	0.60	0.59	0.61
Panel C: Intergenerational persistence			
Ag. employment share, ag-born	0.533	0.472	0.323
Ag. employment share, nonag-born	0.053	0.088	0.114
Sector-choice odds ratio	20.541	9.266	3.717

Note: The Baseline and Added FE columns reproduce the corresponding hourly-earnings estimates in Table 3. The Movers column applies the Added FE procedure to workers whose adult and childhood commuting zones differ, using mover-specific earnings and sector-choice moments and the Added FE estimate of γ. Added FE and Movers adjust the earnings means and sector-choice probabilities for adult commuting-zone and age fixed effects. Earnings moments use workers with valid positive hourly earnings; sector-choice moments entering κ_j, τ, and Panel C use the full linked sample for the relevant column.

A.2.4. Alternative Skill Distributions

Table 10 compares the baseline Fréchet specification with a bivariate lognormal distribution and a bivariate Fréchet distribution joined by a Frank copula. In every specification, the bivariate object is the pair of agricultural and nonagricultural skills, (ϵ_A, ϵ_N), rather than skills indexed by parental origin. Ag-born and nonag-born workers draw from the same joint distribution; parental origin affects sectoral payoffs only through κ_A, κ_N, and τ.

The baseline specification assumes independent unit-scale Fréchet skills with a common shape parameter:

$$F_j(\epsilon) \equiv \Pr(\epsilon_j \leq \epsilon) = \exp\left(-\epsilon^{-\gamma}\right), \qquad \epsilon > 0, \quad j \in \{A, N\}, \qquad \epsilon_A \perp \epsilon_N. \tag{13}$$

The bivariate lognormal specification instead assumes

$$\begin{pmatrix} \log \epsilon_A \\ \log \epsilon_N \end{pmatrix} \sim \mathcal{N}\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_A^2 & \rho\sigma_A\sigma_N \\ \rho\sigma_A\sigma_N & \sigma_N^2 \end{pmatrix} \right]. \tag{14}$$

Finally, the Fréchet–Frank specification allows sector-specific Fréchet marginals,

$$F_j(\epsilon) = \exp\left(-\epsilon^{-\gamma_j}\right), \qquad j \in \{A, N\}, \tag{15}$$

and joins them using the Frank copula

$$\begin{aligned} \Pr(\epsilon_A \leq a, \epsilon_N \leq n) &= C_{\theta_F}(F_A(a), F_N(n)), \\ C_\theta(u, v) &= -\frac{1}{\theta} \log\left[1 + \frac{(e^{-\theta u} - 1)(e^{-\theta v} - 1)}{e^{-\theta} - 1}\right]. \end{aligned} \tag{16}$$

for $\theta \neq 0$, with $C_0(u, v) = uv$ denoting independence. All marginal location or scale parameters are normalized because the relative sector payoff R absorbs their relative level.

I estimate every specification by simulated method of moments using the same seven targets. These consist of the pooled within-cell variance of log earnings, the agricultural employment shares of ag-born and nonag-born workers, the two within-sector parental-origin earnings ratios, and the variances of log earnings among ag-born and nonag-born workers employed in nonagriculture. The five-parameter baseline Fréchet model is overidentified, while the two bivariate specifications each have seven parameters and are exactly identified by these moments.

Table 10: SMM Estimates under Alternative Skill Distributions

	Fréchet	Bivariate lognormal	Fréchet–Frank
Panel A: Skill distribution			
Common shape, γ	2.097		
Agricultural log-skill dispersion, σ_A		0.872	
Nonagricultural log-skill dispersion, σ_N		0.604	
Log-skill correlation, ρ		0.724	
Agricultural shape, γ_A			1.438
Nonagricultural shape, γ_N			2.244
Frank copula parameter, θ_F			1.599
Panel B: Inherited wedges			
Inherited ag. productivity, κ_A	1.586	1.122	2.719
Inherited nonag. productivity, κ_N	1.800	1.234	1.611
Inherited immobility wedge, τ	0.325	0.503	0.253
Relative sector payoff, R	2.204	2.146	3.614
Panel C: Estimation			
SMM objective	0.080	0.085	0.036
Targeted moments	7	7	7
Converged	Yes	Yes	Yes

Note: All specifications minimize an identity-weighted quadratic form in the same seven log moment deviations using fixed common random numbers. The Fréchet column imposes a common shape parameter and independence across sectoral skills. The bivariate lognormal allows sector-specific log-skill dispersions and correlation ρ; the Fréchet–Frank specification allows sector-specific shape parameters and dependence governed by the Frank copula parameter θ_F. The relative sector payoff R is a nuisance parameter. Reported results use 200,000 simulation draws and multiple starting values.

A.2.5. Allocation-Based Illustration of Proposition 1

As a complementary illustration of Proposition 1, I estimate the response of aggregate sectoral allocation to relative earnings using 1940–1950 commuting-zone-by-industry variation. For broad nonagricultural industries j, I estimate

$$\Delta \log \left(\frac{\phi_{j,r}}{\phi_{A,r}} \right) = \beta \Delta \log \left(\frac{y_{j,r}}{y_{A,r}} \right) + \alpha_{j,d(r)} + u_{j,r},$$

where r indexes commuting zones and $\alpha_{j,d(r)}$ denotes industry-by-Census-division fixed effects. The sample stacks seven broad nonagricultural industries, while agriculture is the common reference sector. I instrument relative earnings growth with a leave-one-

commuting-zone-out shift-share measure,

$$Z_{j,r} = s_{j,r,1940} g_{j,-r} - s_{A,r,1940} g_{A,-r},$$

where $s_{j,r,1940}$ is industry j's initial CZ employment share and $g_{j,-r}$ is national 1940–1950 employment growth excluding CZ r. The instrument therefore varies across both commuting zones and industries rather than assigning every industry within a CZ the same shock.

Table 11 reports the resulting aggregate elasticities and applies Proposition 1 using the observed 1940 and 1950 parental composition and subgroup agricultural shares. The population-weighted estimate is $\widehat{\beta} = 1.067$. The corresponding implied within-group elasticity is 1.533 using the 1940 shares and 1.422 using the 1950 shares, approximately one-third above the unadjusted aggregate elasticity. The unweighted estimates imply a smaller within-group elasticity near one. Because the regression stacks comparisons of each nonagricultural industry with agriculture, whereas the proposition is exact for the aggregate two-sector margin, I interpret this exercise as an illustration of the attenuation mechanism rather than as an alternative preferred estimator of γ.

Table 11: Aggregate Reallocation Elasticity Estimates

	(1) Unweighted	(2) Population weighted
Aggregate elasticity $\widehat{\beta}$	0.716	1.067
	(0.284)	(0.334)
First-stage F-statistic	24.21	30.17
Industry–CZ observations	2,423	2,423
CZ clusters	536	536
Industry $\times$ division FE	Yes	Yes

Note: The dependent variable is the 1940–1950 change in log employment share in nonagricultural industry j relative to agriculture. Relative mean hourly earnings growth is instrumented with the leave-one-CZ-out shift-share measure defined above. Employment shares and national employment growth use full-count records, while 1950 earnings moments apply the Census sample-line weights. Both columns include industry-by-Census-division fixed effects; column 2 weights observations by the CZ population of U.S.-born men ages 20–29 in 1940. Standard errors, clustered by commuting zone, are reported in parentheses.

A.2.6. Adjustment for Topcoded Earnings

The main text retains the variance-based estimator but checks its sensitivity to replacing each topcoded annual income with a multiple of the $5,000 censoring threshold. In the final variance-estimation sample, 2,164 of 2,077,253 workers (0.104 percent) are topcoded, and their capped observations account for 0.501 percent of total wage and salary income. I calibrate the multiplier to 2.926 so that these workers instead account for 1.452 percent of total income, the 1940 income share of the top 0.1 percent reported by Atkinson, Piketty and Saez (2011). I then reconstruct hourly earnings and reestimate γ from the residual variance on the fixed main-text sample.

Table 12 shows that the estimate changes little over a broad range of multipliers. At the calibrated value, $\widehat{\gamma}$ declines from 2.076 to 2.063, while the quantile RMSE rises only from 0.337 to 0.340.

Table 12: Sensitivity to the Topcode Adjustment

Topcode multiplier	Topcoded income share (%)	$\widehat{\gamma}$	Observations	Quantile RMSE
1.00	0.501	2.076	2,077,253	0.337
1.25	0.625	2.074	2,077,253	0.338
1.50	0.749	2.072	2,077,253	0.338
1.75	0.873	2.070	2,077,253	0.339
2.00	0.997	2.068	2,077,253	0.339
2.93	1.452	2.063	2,077,253	0.340

Note: Each row replaces topcoded annual income with the stated multiple of $5,000 before constructing hourly earnings and reestimating γ. The 2.93 multiplier matches the 1.452 percent target share; quantile RMSE is measured over the 1st–99th residual percentiles.

A.2.7. Visualization of Empirical vs Model Distribution

For a visual check, Figure 13 compares the residual earnings distribution with the log-Fréchet densities implied by the baseline and topcode-adjusted estimates.

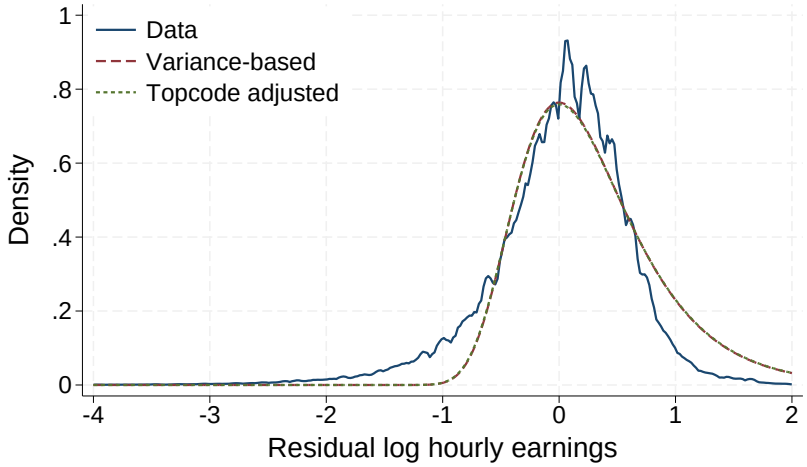


Figure 13: Residual Earnings Distribution in Data and Model

Notes: The solid line plots the kernel density of residual log hourly wage and salary income after removing worker-sector-by-parental-sector cell means. The dashed red and short-dashed green lines use the baseline and topcode-adjusted estimates, $\widehat{\gamma} = 2.076$ and 2.063, respectively. The estimation sample has positive wage and salary income, positive weeks worked, and at least 35 usual weekly hours, with no log-earnings restriction; the figure displays residuals between -4 and 2.

A.2.8. Alternative Earnings Measures

Hourly earnings are the closest empirical counterpart to the model's wage per efficiency unit, but they inherit measurement error from both weeks and hours. Annual and weekly earnings provide useful checks on whether this additional measurement affects the estimated dispersion of earnings. For comparability, Table 13 estimates all three concepts on the common preferred sample used in the main text.

Table 13: Variance-Based Estimates of the Fréchet Shape Parameter by Income Measure

	Annual		Weekly		Hourly	
	Baseline	Added FE	Baseline	Added FE	Baseline	Added FE
$\widehat{\gamma}$	2.187	2.414	2.214	2.441	2.076	2.288
Residual SD of log wages	0.586	0.531	0.579	0.525	0.618	0.561
Observations	2,077,253	2,077,253	2,077,253	2,077,253	2,077,253	2,077,253
Topcoded observations	2,164	2,164	2,164	2,164	2,164	2,164
Topcoded share (%)	0.104	0.104	0.104	0.104	0.104	0.104
Earnings regression R^2	0.307	0.431	0.306	0.430	0.341	0.458

Note: Estimates use $\widehat{\gamma} = \pi/(\sqrt{6}\widehat{\sigma})$ after removing worker-sector-by-parental-sector effects; added-FE columns also control for adult commuting zone and age. Weekly earnings divide annual wage and salary income by weeks worked, and hourly earnings additionally divide by usual weekly hours. All columns use the common unrestricted 2,077,253-observation sample defined in Table 2.

The estimates are similar across earnings concepts: the baseline estimates range from 2.076 for hourly earnings to 2.214 for weekly earnings, while the added-fixed-effect estimates range from 2.288 to 2.441. Figure 14 shows the corresponding distributional fits.

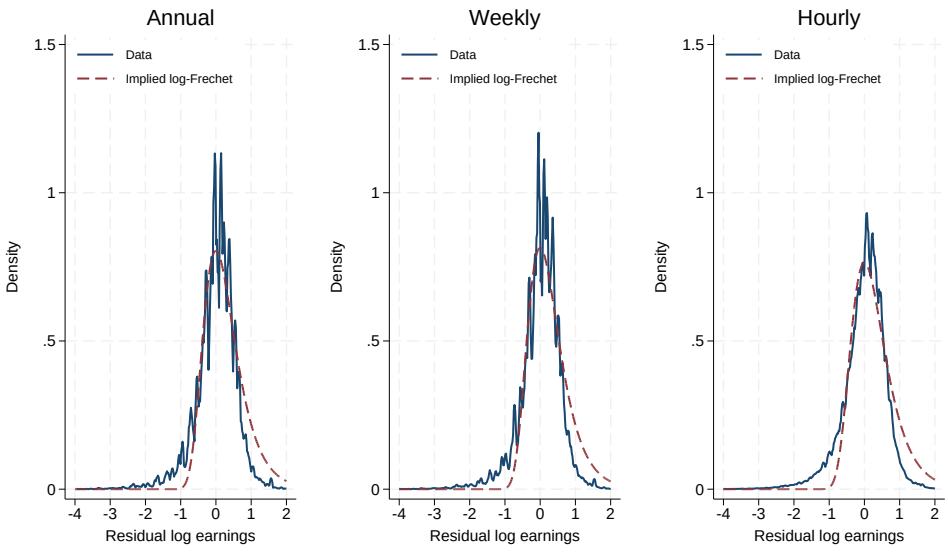


Figure 14: Residual Earnings Distributions and Implied Log-Fréchet Densities by Income Measure

Note: The solid lines plot kernel densities of residual log earnings after removing worker-sector-by-parental-sector cell means. Dashed lines plot the log-Fréchet densities implied by the baseline estimates in Table 13. Panels use the common unrestricted estimation sample and display residuals between -4 and 2.

Finally, Table 14 carries each variance-based estimate of γ through the wedge calculations. The estimated inherited productivity and immobility wedges remain economically important across earnings concepts and control sets, although their magnitudes vary with the dispersion estimate and adjusted income gaps.

Table 14: Inherited Productivity and Immobility Wedges by Income Measure

	Annual earnings		Weekly earnings		Hourly earnings	
	Baseline	Added FE	Baseline	Added FE	Baseline	Added FE
Panel A: Inferred wedges						
Inherited agricultural productivity, $\widehat{\kappa}_A$	1.939	1.635	1.766	1.492	1.604	1.323
Inherited nonagricultural productivity, $\widehat{\kappa}_N$	1.805	1.485	1.749	1.431	1.797	1.437
Inherited mobility wedge, $\widehat{\tau}$	0.121	0.034	0.211	0.142	0.328	0.281
Panel B: Mean earnings						
Agriculture, ag-born: $\widehat{y}_A(A)$	329.62	347.82	8.68	9.19	0.19	0.20
Agriculture, nonag-born: $\widehat{y}_A(N)$	489.81	426.55	13.99	12.25	0.37	0.32
Nonagriculture, ag-born: $\widehat{y}_N(A)$	700.11	758.24	17.58	19.14	0.47	0.52
Nonagriculture, nonag-born: $\widehat{y}_N(N)$	914.29	897.82	22.33	21.89	0.60	0.59
Panel C: Intergenerational persistence						
Sector-choice odds ratio	20.541	9.266	20.541	9.266	20.541	9.266

Note: Each column uses the corresponding estimate of γ in Table 13. Added-FE columns adjust income and sector-choice moments for adult commuting zone and age. Earnings moments use the relevant income sample; all sector-choice moments entering κ_j, τ, and Panel C use the full linked sample, as in Table 3.

B. Theoretical Results

This appendix collects the proofs and supporting derivations for the theoretical results in Sections 2 and 3.

B.1. Aggregate Reallocation in Partial Equilibrium

Proof of Proposition 1. Holding parental composition and the relative payoff shifter fixed, equation (1) implies

$$\frac{\partial \phi_{A,t}(k)}{\partial \log(w_{N,t}/w_{A,t})} = -\gamma \phi_{A,t}(k)\phi_{N,t}(k).$$

Aggregating across parental groups therefore gives

$$\frac{\partial \phi_{A,t}}{\partial \log(w_{N,t}/w_{A,t})} = -\gamma \sum_k \phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k).$$

The aggregate relative-allocation elasticity is therefore

$$\mathcal{E} = \gamma \frac{\sum_k \phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)}{\phi_{A,t}\phi_{N,t}}.$$

For the agricultural-choice indicator $D_{i,t}$, the law of total variance gives

$$\mathrm{Var}(D_{i,t}) = \mathbb{E}[\mathrm{Var}(D_{i,t} \mid k)] + \mathrm{Var}(\mathbb{E}[D_{i,t} \mid k]),$$

where

$$\mathrm{Var}(D_{i,t}) = \phi_{A,t}\phi_{N,t}, \qquad \mathbb{E}[\mathrm{Var}(D_{i,t} \mid k)] = \sum_k \phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k).$$

Substitution yields

$$\mathcal{E} = \gamma \left[1 - \frac{\mathrm{Var}(\mathbb{E}[D_{i,t} \mid k])}{\mathrm{Var}(D_{i,t})}\right].$$

The within-group Bernoulli variances are strictly positive, so $\mathcal{E} > 0$. The variance ratio is nonnegative, implying $\mathcal{E} \leq \gamma$. The inequality is strict when both parental groups have positive mass and $\phi_{A,t}(A) \neq \phi_{A,t}(N)$; otherwise the between-group variance is zero and $\mathcal{E} = \gamma$. □

B.2. Contemporaneous Transmission of Agricultural Technology Shock

The main text reduces the contemporaneous equilibrium to a relative headcount labor-supply schedule,

$$d\log\frac{\phi_{N,t}}{\phi_{A,t}} = \mathcal{E}\, d\log\frac{w_{N,t}}{w_{A,t}},$$

and an inverse relative labor-demand schedule expressed in terms of headcount allocations,

$$d\log\frac{w_{N,t}}{w_{A,t}} = \mathcal{D}^{Z_A} d\log Z_{A,t} + \mathcal{H}\, d\log\frac{\phi_{N,t}}{\phi_{A,t}}.$$

The three objects in these schedules have economically distinct ceteris-paribus interpretations. The technology elasticity $\mathcal{D}^{Z_A}$ shifts inverse relative labor demand while holding sectoral effective labor supplies fixed:

$$\mathcal{D}^{Z_A} \equiv \left.\frac{\partial \log(w_{N,t}/w_{A,t})}{\partial \log Z_{A,t}}\right|_{d\log H_{A,t} = d\log H_{N,t} = 0}.$$

The worker-allocation elasticity $\mathcal{E}$ holds cohort composition and the relative payoff shifter fixed. The feedback elasticity $\mathcal{H}$ holds technologies fixed and traces how effective la-

bor, production, and relative goods prices respond along the same within-cohort worker-allocation margin underlying $\mathcal{E}$. Solving the two schedules and using $d\phi_{A,t} = -\phi_{A,t}(1-\phi_{A,t})d\log(\phi_{N,t}/\phi_{A,t})$ gives

$$\frac{d\phi_{A,t}}{d\log Z_{A,t}} = -\phi_{A,t}(1-\phi_{A,t})\frac{\mathcal{E}}{1-\mathcal{E}\mathcal{H}}\mathcal{D}^{Z_A}.$$

Below I derive each schedule from the full equilibrium system. I begin with the benchmark version of the economy without intergenerational persistence.

Supporting Derivations without IGP The agricultural technology innovation provides a direct shock to labor demand:

$$d\log\frac{w_{N,t}}{w_{A,t}} \quad = \quad d\log Z_{N,t} \quad - \quad d\log Z_{A,t} \quad + \quad d\log p_t$$

It also provides an indirect shock to labor demand through the relative price:

$$d\log p_t \quad = \quad -\frac{1}{\sigma}\,d\log\frac{Q_{N,t}}{Q_{A,t}} \quad + \quad \frac{(\vartheta_N-\vartheta_A)(1-\sigma)}{\sigma}\,d\log Q_t$$

given both the relative sectoral and aggregate output respond to the agricultural technology shock:

$$\begin{aligned}
d\log Q_{A,t} &= d\log Z_{A,t} + d\log H_{A,t} \\
d\log Q_{N,t} &= d\log Z_{N,t} + d\log H_{N,t} \\
d\log Q_t &= \frac{\omega_{A,t}}{\omega_{A,t}\vartheta_A+\omega_{N,t}\vartheta_N}\,d\log Q_{A,t} + \frac{\omega_{N,t}}{\omega_{A,t}\vartheta_A+\omega_{N,t}\vartheta_N}\,d\log Q_{N,t}
\end{aligned}$$

Hence, holding the effective labor supply fixed, the contemporaneous response of the relative wage to the agricultural technology shock is given by:

$$\mathcal{D}^{Z_A} \quad \equiv \quad \left.\frac{d\log(w_{N,t}/w_{A,t})}{d\log Z_{A,t}}\right|_{dH_{j,t}=0} \quad = \quad \frac{1-\sigma}{\sigma}\cdot\frac{\vartheta_N}{\omega_{A,t}\vartheta_A+\omega_{N,t}\vartheta_N}$$

The relative worker allocation satisfies:

$$d\log\frac{\phi_{N,t}}{\phi_{A,t}} \quad = \quad \gamma\cdot\left[d\log\frac{w_{N,t}}{w_{A,t}} \quad + \quad d\log\frac{B_{N,t}}{B_{A,t}}\right]$$

and the effective labor supply satisfies:

$$d\log H_{A,t} = \frac{\gamma-1}{\gamma}\cdot d\log\phi_{A,t} \qquad \text{and} \qquad d\log H_{N,t} = -\frac{\phi_{A,t}}{1-\phi_{A,t}}\cdot d\log H_{A,t}$$

Substituting these into the relative output equation yields:

$$d\log\frac{Q_{N,t}}{Q_{A,t}} = -\frac{1}{1-\phi_{A,t}}\cdot\frac{\gamma-1}{\gamma}\cdot d\log\phi_{A,t}$$

while substituting into the aggregate output equation yields:

$$d\log Q_t = \left[\frac{\omega_{A,t}-\phi_{A,t}}{(1-\phi_{A,t})\left(\omega_{A,t}\vartheta_A+\omega_{N,t}\vartheta_N\right)}\right]\cdot\frac{\gamma-1}{\gamma}\cdot d\log\phi_{A,t}$$

Hence, the relative wage responds to worker reallocation according to:

$$d\log\frac{w_{N,t}}{w_{A,t}} = -\frac{\gamma-1}{\gamma}\cdot\left[\frac{1}{\sigma} + \frac{(\vartheta_N-\vartheta_A)(1-\sigma)}{\sigma}\cdot\frac{\omega_{A,t}-\phi_{A,t}}{\omega_{A,t}\vartheta_A+\omega_{N,t}\vartheta_N}\right]\cdot d\log\frac{\phi_{N,t}}{\phi_{A,t}}$$

Define:

$$\mathcal{E} \equiv \left.\frac{\partial\log(\phi_{N,t}/\phi_{A,t})}{\partial\log(w_{N,t}/w_{A,t})}\right|_{d\log(B_{N,t}/B_{A,t})=0} = \gamma.$$

Similarly,

$$\begin{aligned}\mathcal{H} &\equiv \left.\frac{d\log(w_{N,t}/w_{A,t})}{d\log(\phi_{N,t}/\phi_{A,t})}\right|_{d\log Z_{A,t}=d\log Z_{N,t}=0}\\ &= -\frac{\gamma-1}{\gamma}\cdot\left[\frac{1}{\sigma} + \frac{(\vartheta_N-\vartheta_A)(1-\sigma)}{\sigma}\cdot\frac{\omega_{A,t}-\phi_{A,t}}{\omega_{A,t}\vartheta_A+\omega_{N,t}\vartheta_N}\right]\end{aligned}$$

Then, the relative worker allocation response to a labor demand shock originating from the agricultural technology shock is given by:

$$\begin{aligned}d\log\frac{\phi_{N,t}}{\phi_{A,t}} &= \mathcal{E}\cdot d\log\frac{w_{N,t}}{w_{A,t}}\\ &= \mathcal{E}\cdot\left[\mathcal{D}^{Z_A}\,d\log Z_{A,t}+\mathcal{H}\cdot d\log\frac{\phi_{N,t}}{\phi_{A,t}}\right]\\ &= \frac{\mathcal{E}}{1-\mathcal{E}\cdot\mathcal{H}}\cdot\mathcal{D}^{Z_A}\cdot d\log Z_{A,t}\end{aligned}$$

Hence, as noted above, I arrive at:

$$\frac{d\phi_{A,t}}{d\log Z_{A,t}} \quad = \quad -\phi_{At}(1-\phi_{A,t}) \cdot \frac{\mathcal{E}}{1-\mathcal{E}\cdot\mathcal{H}} \cdot \mathcal{D}^{Z_A}$$

This expression will hold even in the presence of intergenerational persistence, but the underlying closed-form solutions for $\mathcal{E}$ and $\mathcal{H}$ will change.

Worker-allocation elasticity with IGP Define

$$S_{0,t} \equiv \sum_{k\in\{A,N\}} \phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k).$$

Equation (1) implies

$$\frac{\partial\phi_{A,t}(k)}{\partial\log(w_{N,t}/w_{A,t})} = -\gamma\phi_{A,t}(k)\phi_{N,t}(k).$$

Holding cohort composition fixed and aggregating across parental sectors therefore gives

$$\frac{\partial\phi_{A,t}}{\partial\log(w_{N,t}/w_{A,t})} = -\gamma S_{0,t}.$$

Because $\phi_{N,t} = 1-\phi_{A,t}$,

$$\mathcal{E} = \gamma\frac{S_{0,t}}{\phi_{A,t}\phi_{N,t}}.$$

Under stationarity, $\phi_{A,t-1} = \phi_{A,t}$, equation (3) implies

$$\frac{S_{0,t}}{\phi_{A,t}\phi_{N,t}} = 1-\left[\phi_{A,t}(A)-\phi_{A,t}(N)\right]^2.$$

Thus the worker-allocation elasticity in an economy with IGP is

$$\mathcal{E} = \gamma\left\{1-\left[\phi_{A,t}(A)-\phi_{A,t}(N)\right]^2\right\}.$$

General-equilibrium feedback with IGP To derive $\mathcal{H}$ without imposing homothetic demand, first define the normalized sector-specific effective-labor responses

$$\Psi_{A,t} \equiv \frac{\phi_{A,t}\phi_{N,t}}{S_{0,t}}\sum_{k\in\{A,N\}}\phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)\frac{h_{A,t}(k)}{H_{A,t}},$$
$$\Psi_{N,t} \equiv \frac{\phi_{A,t}\phi_{N,t}}{S_{0,t}}\sum_{k\in\{A,N\}}\phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)\frac{h_{N,t}(k)}{H_{N,t}}.$$

Differentiating equation (2) gives

$$d\log h_{j,t}(k) = -\frac{1}{\gamma} d\log \phi_{j,t}(k).$$

Combining this selection response with the subcohort allocation response gives

$$\frac{\partial \log H_{A,t}}{\partial \log(w_{N,t}/w_{A,t})} = -(\gamma-1)\sum_k \phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)\frac{h_{A,t}(k)}{H_{A,t}},$$
$$\frac{\partial \log H_{N,t}}{\partial \log(w_{N,t}/w_{A,t})} = (\gamma-1)\sum_k \phi_{k,t-1}\phi_{A,t}(k)\phi_{N,t}(k)\frac{h_{N,t}(k)}{H_{N,t}}.$$

Dividing by the worker-allocation elasticity derived above yields

$$\frac{d\log H_{A,t}}{d\log(\phi_{N,t}/\phi_{A,t})} = -\frac{\gamma-1}{\gamma}\Psi_{A,t},$$
$$\frac{d\log H_{N,t}}{d\log(\phi_{N,t}/\phi_{A,t})} = \frac{\gamma-1}{\gamma}\Psi_{N,t}.$$

Holding technologies fixed, sectoral production implies $d\log Q_{j,t} = d\log H_{j,t}$. Hence

$$\frac{d\log(Q_{N,t}/Q_{A,t})}{d\log(\phi_{N,t}/\phi_{A,t})} = \frac{\gamma-1}{\gamma}(\Psi_{N,t}+\Psi_{A,t}),$$

and the final-good aggregator implies

$$\frac{d\log Q_t}{d\log(\phi_{N,t}/\phi_{A,t})} = \frac{\gamma-1}{\gamma}\frac{\omega_{N,t}\Psi_{N,t}-\omega_{A,t}\Psi_{A,t}}{\omega_{A,t}\vartheta_A+\omega_{N,t}\vartheta_N}.$$

Substitution into the relative-wage equation gives the full nonhomothetic feedback elasticity:

$$\mathcal{H} = \frac{\gamma-1}{\gamma}\left\{-\frac{\Psi_{A,t}+\Psi_{N,t}}{\sigma} + \frac{(\vartheta_N-\vartheta_A)(1-\sigma)}{\sigma}\frac{\omega_{N,t}\Psi_{N,t}-\omega_{A,t}\Psi_{A,t}}{\omega_{A,t}\vartheta_A+\omega_{N,t}\vartheta_N}\right\}.$$

The first term is the relative-price feedback; the second is the aggregate-income feedback generated by nonhomothetic demand. In the homothetic case, $\vartheta_A = \vartheta_N = 1$, the second term vanishes and

$$\mathcal{H} = -\frac{1}{\sigma}\frac{\gamma-1}{\gamma}(\Psi_{A,t}+\Psi_{N,t}).$$

In the benchmark economy, $\Psi_{A,t} = \phi_{N,t}$ and $\Psi_{N,t} = \phi_{A,t}$. Thus the nonhomothetic expression nests the earlier benchmark result, while its homothetic specialization reduces

to $\mathcal{H}^{\text{bench}} = -(\gamma - 1)/(\gamma\sigma)$.

Application of Proposition 1 to the Stylized Economies Proposition 1 implies:

> The elasticity of worker allocation with respect to the relative wage is strictly smaller in the presence of intergenerational persistence regardless of whether that persistence arises from inherited productivity or inherited immobility.

$$0 \quad < \quad \mathcal{E}^{\text{IGP-}\kappa} \quad = \quad \mathcal{E}^{\text{IGP-}\tau} \quad < \quad \mathcal{E}^{\text{bench}} \quad = \quad \gamma$$

The stationary expression for $\mathcal{E}$ shows that it depends on IGP only through $\phi_{A,t}(A) - \phi_{A,t}(N)$. In the benchmark economy, $\phi_{A,t}(A) - \phi_{A,t}(N) = 0$, so $\mathcal{E}^{\text{bench}} = \gamma$. Each IGP economy has $0 < \phi_{A,t}(A) - \phi_{A,t}(N) < 1$, implying

$$0 < \gamma \left\{1 - [\phi_{A,t}(A) - \phi_{A,t}(N)]^2\right\} < \gamma.$$

Finally, the two IGP economies are constructed to have the same allocation gap and therefore have the same elasticity.

Proof of Lemma 1 Recall, Lemma 1 states:

> The magnitude of the general-equilibrium feedback of worker reallocation into the relative wage is strictly larger in the presence of intergenerational persistence, and it is largest when that persistence arises from inherited immobility:
>
> $$\mathcal{H}^{\text{IGP-}\tau} < \mathcal{H}^{\text{IGP-}\kappa} \leq \mathcal{H}^{\text{bench}} = -\frac{1}{\sigma}\frac{\gamma - 1}{\gamma},$$
>
> where the weak inequality is an equality only when $\phi_{A,t-1} = 0.5$.

Proof. Under homothetic demand, the expression for $\mathcal{H}$ above shows that its ordering is the reverse of the ordering of $\Psi_{A,t} + \Psi_{N,t}$. The benchmark values $\Psi_{A,t} = \phi_{N,t}$ and $\Psi_{N,t} = \phi_{A,t}$ imply $\Psi_{A,t} + \Psi_{N,t} = 1$ and hence

$$\mathcal{H}^{\text{bench}} = -\frac{1}{\sigma}\frac{\gamma - 1}{\gamma}.$$

For the two IGP economies, equation (1) gives

$$\frac{\phi_{A,t}(A)/\phi_{N,t}(A)}{\phi_{A,t}(N)/\phi_{N,t}(N)} = \begin{cases} \kappa^{2\gamma} & \text{in the IGP-}\kappa\text{ economy} \\ (1-\tau)^{-\gamma} & \text{in the IGP-}\tau\text{ economy.} \end{cases}$$

The common allocation gap and common stationary aggregate allocation therefore pin down the wedges being compared, but the ordering does not follow from the odds ratio alone. Stationarity implies

$$\phi_{A,t}(A) = \phi_{A,t} + \phi_{N,t}[\phi_{A,t}(A) - \phi_{A,t}(N)], \qquad \phi_{A,t}(N) = \phi_{A,t}\{1 - [\phi_{A,t}(A) - \phi_{A,t}(N)]\}.$$

The odds-ratio restriction for the IGP-κ economy becomes

$$\kappa^{2\gamma} = \frac{\phi_{A,t}(A)\phi_{N,t}(N)}{\phi_{N,t}(A)\phi_{A,t}(N)}.$$

Substituting equation (2) into the definitions of $\Psi_{A,t}$ and $\Psi_{N,t}$ gives

$$\Psi_{A,t}(\kappa) = \frac{1}{1-[\phi_{A,t}(A)-\phi_{A,t}(N)]^2}\frac{\phi_{A,t}\kappa\phi_{A,t}(A)^{(\gamma-1)/\gamma}\phi_{N,t}(A) + \phi_{N,t}\phi_{A,t}(N)^{(\gamma-1)/\gamma}\phi_{N,t}(N)}{\phi_{A,t}\kappa\phi_{A,t}(A)^{(\gamma-1)/\gamma} + \phi_{N,t}\phi_{A,t}(N)^{(\gamma-1)/\gamma}},$$
$$\Psi_{N,t}(\kappa) = \frac{1}{1-[\phi_{A,t}(A)-\phi_{A,t}(N)]^2}\frac{\phi_{A,t}\phi_{A,t}(A)\phi_{N,t}(A)^{(\gamma-1)/\gamma} + \phi_{N,t}\kappa\phi_{A,t}(N)\phi_{N,t}(N)^{(\gamma-1)/\gamma}}{\phi_{A,t}\phi_{N,t}(A)^{(\gamma-1)/\gamma} + \phi_{N,t}\kappa\phi_{N,t}(N)^{(\gamma-1)/\gamma}}.$$

The IGP-τ expressions are the same formulas evaluated at $\kappa = 1$, because inherited immobility affects the common subcohort allocations but does not directly augment effective labor.

Since $\phi_{A,t}(A) > \phi_{A,t}(N)$, we have $\phi_{N,t}(A) < \phi_{N,t}(N)$. The fraction defining $\Psi_{A,t}(\kappa)$ is a weighted average of $\phi_{N,t}(A)$ and $\phi_{N,t}(N)$; raising κ places more weight on the smaller value $\phi_{N,t}(A)$. Similarly, the fraction defining $\Psi_{N,t}(\kappa)$ is a weighted average of $\phi_{A,t}(A)$ and $\phi_{A,t}(N)$; raising κ places more weight on the smaller value $\phi_{A,t}(N)$. Because $\kappa > 1$, both comparisons are strict:

$$\Psi_{A,t}^{\text{IGP-}\tau} > \Psi_{A,t}^{\text{IGP-}\kappa}, \qquad \Psi_{N,t}^{\text{IGP-}\tau} > \Psi_{N,t}^{\text{IGP-}\kappa}.$$

For the comparison with the benchmark, substitute the odds restriction into the preceding expressions for $\Psi_{A,t}(\kappa)$ and $\Psi_{N,t}(\kappa)$ and put the two sectoral terms over a common denominator. After cancellation, the sign of $\Psi_{A,t}^{\text{IGP-}\kappa} + \Psi_{N,t}^{\text{IGP-}\kappa} - 1$ is the sign of $(2\phi_{A,t} - 1)^2$.

Consequently,

$$(\Psi_{A,t} + \Psi_{N,t})^{\text{IGP-}\tau} > (\Psi_{A,t} + \Psi_{N,t})^{\text{IGP-}\kappa} \geq 1.$$

The second inequality is strict unless $\phi_{A,t} = 0.5$; at that allocation, the two symmetric inherited-productivity terms offset exactly. Multiplication by the negative factor $-(\gamma - 1)/(\gamma\sigma)$ in the homothetic expression for $\mathcal{H}$ reverses the inequalities and gives the stated ordering. □

Proof of Proposition 2 Recall, Proposition 2 states:

> Improvements in agricultural technology induce a reallocation of workers out of agriculture, but the magnitude of that reallocation is smaller in the presence of intergenerational persistence:

$$\left(\frac{d\phi_{A,t}}{d\log Z_{A,t}}\right)^{\text{bench}} < \left(\frac{d\phi_{A,t}}{d\log Z_{A,t}}\right)^{\text{IGP-}\kappa} < \left(\frac{d\phi_{A,t}}{d\log Z_{A,t}}\right)^{\text{IGP-}\tau} < 0.$$

Proof. The definitions of $\mathcal{D}^{Z_A}$, $\mathcal{E}$, and $\mathcal{H}$ imply:

$$\frac{d\log(\phi_{N,t}/\phi_{A,t})}{d\log Z_{A,t}} = \mathcal{E}\cdot\left[\mathcal{D}^{Z_A} + \mathcal{H}\cdot\frac{d\log(\phi_{N,t}/\phi_{A,t})}{d\log Z_{A,t}}\right].$$

Solving this expression and using $d\phi_{A,t} = -\phi_{A,t}(1-\phi_{A,t})d\log(\phi_{N,t}/\phi_{A,t})$ gives:

$$\frac{d\phi_{A,t}}{d\log Z_{A,t}} = -\phi_{A,t}(1-\phi_{A,t})\cdot\frac{\mathcal{E}}{1-\mathcal{E}\mathcal{H}}\cdot\mathcal{D}^{Z_A}.$$

The response is negative because $\mathcal{D}^{Z_A} > 0$, $\mathcal{E} > 0$, and $\mathcal{H} < 0$. The three economies have the same initial aggregate allocation and the same $\mathcal{D}^{Z_A}$, so their responses can be ordered by $\mathcal{E}/(1-\mathcal{E}\mathcal{H})$. This term can be written as:

$$\frac{\mathcal{E}}{1-\mathcal{E}\mathcal{H}} = \frac{1}{1/\mathcal{E}-\mathcal{H}}.$$

Proposition 1, together with the common allocation-gap restriction, implies that $1/\mathcal{E}$ is smallest in the benchmark economy and equal across the two IGP economies, while Lemma 1 implies that $-\mathcal{H}$ is smallest in the benchmark economy and largest in the IGP-τ economy. Therefore:

$$\left(\frac{\mathcal{E}}{1-\mathcal{E}\mathcal{H}}\right)^{\text{bench}} > \left(\frac{\mathcal{E}}{1-\mathcal{E}\mathcal{H}}\right)^{\text{IGP-}\kappa} > \left(\frac{\mathcal{E}}{1-\mathcal{E}\mathcal{H}}\right)^{\text{IGP-}\tau} > 0.$$

Multiplying by the common negative term $-\phi_{A,t}(1-\phi_{A,t})\mathcal{D}^{Z_A}$ reverses these inequalities and proves Proposition 2. □

B.3. Dynamic Transmission of Past Reallocation via Parent Composition

With intergenerational persistence, past worker reallocation can affect future worker allocation through the composition of the parent generation. As in the main text, I separate this composition change into two initial shifts: a direct shift in aggregate headcounts and an indirect shift in the relative wage through effective labor supply. The direct shift follows from the aggregate worker allocation in the next period:

$$\phi_{A,t+1} \quad = \quad \phi_{A,t+1}(A)\,\phi_{A,t} \quad + \quad \phi_{A,t+1}(N)\,\phi_{N,t}$$

Hence, I have:

$$\mathcal{F}^{direct} \quad \equiv \quad \left.\frac{\partial \phi_{A,t+1}}{\partial \phi_{A,t}}\right|_{d\phi_{j,t+1}(k)=0} \quad = \quad \phi_{A,t+1}(A) - \phi_{A,t+1}(N).$$

The indirect shift operates through the effective labor supplied to each sector, which in turn affects the relative wage and the next-period subcohort allocations. The indirect effect is given by:

$$\mathcal{F}^{H_A} \equiv \left.\frac{\partial H_{A,t+1}}{\partial \phi_{A,t}}\right|_{d\phi_{j,t+1}(k)=0} = \phi_{A,t+1}(A)h_{A,t+1}(A) - \phi_{A,t+1}(N)h_{A,t+1}(N),$$

$$\mathcal{F}^{H_N} \equiv \left.\frac{\partial H_{N,t+1}}{\partial \phi_{A,t}}\right|_{d\phi_{j,t+1}(k)=0} = \phi_{N,t+1}(A)h_{N,t+1}(A) - \phi_{N,t+1}(N)h_{N,t+1}(N).$$

Thus the composition-induced relative-wage response, holding the period-$t+1$ subcohort allocations fixed, is

$$\begin{aligned}\mathcal{F}^{indirect} &\equiv \left.\frac{\partial \log(w_{N,t+1}/w_{A,t+1})}{\partial \phi_{A,t}}\right|_{d\phi_{j,t+1}(k)=0} \\ &= \frac{1}{\sigma}\left(\frac{\mathcal{F}^{H_A}}{H_{A,t+1}} - \frac{\mathcal{F}^{H_N}}{H_{N,t+1}}\right) \\ &\quad + \frac{(\vartheta_N - \vartheta_A)(1-\sigma)}{\sigma}\frac{\omega_{A,t+1}\mathcal{F}^{H_A}/H_{A,t+1} + \omega_{N,t+1}\mathcal{F}^{H_N}/H_{N,t+1}}{\omega_{A,t+1}\vartheta_A + \omega_{N,t+1}\vartheta_N}.\end{aligned}$$

The first term is the relative-price response to the composition-induced change in relative effective labor. The second term is the corresponding aggregate-income response under nonhomothetic demand.

To incorporate the contemporaneous equilibrium response at $t+1$, let

$$v_{t+1} \equiv \frac{d\log(w_{N,t+1}/w_{A,t+1})}{d\phi_{A,t}}.$$

The dynamic equilibrium can then be reduced to the same two equations used in the main text. Differentiating the aggregate next-period worker allocation gives the headcount response

$$\frac{d\phi_{A,t+1}}{d\phi_{A,t}} = \mathcal{F}^{direct} - \phi_{A,t+1}\phi_{N,t+1}\mathcal{E}v_{t+1},$$

while the relative-wage response satisfies

$$v_{t+1} = \mathcal{F}^{indirect} + \mathcal{H}\mathcal{E}v_{t+1}.$$

The initial composition-induced wage response is $\mathcal{F}^{indirect}$. It induces a within-cohort worker reallocation of $\mathcal{E}v_{t+1}$, which feeds back into the relative wage through $\mathcal{H}$. In contrast, $\mathcal{F}^{direct}$ changes aggregate headcounts by changing cohort weights and therefore does not enter the $\mathcal{E}$–$\mathcal{H}$ feedback loop. Solving the relative-wage equation gives

$$v_{t+1} = \frac{\mathcal{F}^{indirect}}{1-\mathcal{E}\mathcal{H}}.$$

Substituting this response into the headcount equation gives the total dynamic response:

$$\boxed{\frac{d\phi_{A,t+1}}{d\phi_{A,t}} = \mathcal{F}^{direct} - \frac{\phi_{A,t+1}\phi_{N,t+1}\mathcal{E}\mathcal{F}^{indirect}}{1-\mathcal{E}\mathcal{H}}}.$$

The denominator $1-\mathcal{E}\mathcal{H}$ therefore applies only to the worker reallocation initiated by the composition-induced wage response.

Proof of Lemma 2 Recall, Lemma 2 states:

> Holding wages fixed, future generations respond to past reallocation according to
>
> $$0 = \mathcal{F}^{direct,\text{bench}} < \mathcal{F}^{direct,\text{IGP-}\kappa} = \mathcal{F}^{direct,\text{IGP-}\tau}.$$

Proof. In the benchmark economy, equation (1) gives $\phi_{A,t+1}(A) = \phi_{A,t+1}(N)$ and therefore $\mathcal{F}^{direct,\text{bench}} = 0$. In each IGP economy, ag-born workers are more likely to choose agriculture than nonag-born workers, so $\mathcal{F}^{direct} > 0$. The restriction that the two IGP economies have the same allocation gap implies:

$$0 \quad = \quad \mathcal{F}^{direct,\text{bench}} \quad < \quad \mathcal{F}^{direct,\text{IGP-}\kappa} \quad = \quad \mathcal{F}^{direct,\text{IGP-}\tau}.$$

□

Proof of Lemma 3 Recall, Lemma 3 states:

> The relative wage in period $t+1$ responds to past worker reallocation according to
>
> $$\mathcal{F}^{indirect,\text{IGP-}\kappa} > \mathcal{F}^{indirect,\text{IGP-}\tau} > \mathcal{F}^{indirect,\text{bench}} = 0.$$

Proof. In the benchmark economy, parental sector does not affect either sector choice or effective labor supply, so $\mathcal{F}^{H_A} = \mathcal{F}^{H_N} = 0$. Therefore, $\mathcal{F}^{indirect,\text{bench}} = 0$. Impose homothetic demand, $\vartheta_A = \vartheta_N = 1$, as maintained in the statement of the lemma. The aggregate-income term in the expression for $\mathcal{F}^{indirect}$ then vanishes. In either IGP economy, an increase in the fraction of ag-born parents raises effective labor supply in agriculture relative to nonagriculture. The remaining relative-price term therefore implies $\mathcal{F}^{indirect} > 0$. Substituting equation (2) into $\mathcal{F}^{H_A}$ and $\mathcal{F}^{H_N}$ shows that inherited productivity strictly increases the composition-induced effective-labor difference relative to inherited immobility. Since the two IGP economies have the same allocation gap, it follows that:

$$\mathcal{F}^{indirect,\text{IGP-}\kappa} \quad > \quad \mathcal{F}^{indirect,\text{IGP-}\tau} \quad > \quad \mathcal{F}^{indirect,\text{bench}} \quad = \quad 0.$$

□

Proof of Proposition 3 Recall, Proposition 3 states:

> Past worker reallocation propagates positively in the IGP-τ economy and does not propagate in the benchmark economy:
>
> $$\left(\frac{d\phi_{A,t+1}}{d\phi_{A,t}}\right)^{\text{IGP-}\tau} > 0, \qquad \left(\frac{d\phi_{A,t+1}}{d\phi_{A,t}}\right)^{\text{bench}} = 0.$$
>
> In the IGP-κ economy, the derivative can be positive or negative depending

on whether the direct composition effect or the induced relative-wage effect dominates.

The following power-function inequality supplies the algebraic step needed for the inherited-immobility case.

Lemma 4 (Two-state concavity inequality)**.** *Let* $0 < m < 1$, $0 < \delta < 1$, *and* $\gamma > 1$, *and consider four shares that satisfy*

$$a = m + (1-m)\delta, \qquad b = m(1-\delta), \qquad c = 1-a, \qquad e = 1-b.$$

Then

$$\begin{aligned} &m(1-m)(\gamma-\delta^2)\left\{(ae)^{(\gamma-1)/\gamma} - (bc)^{(\gamma-1)/\gamma}\right\} \\ &\quad < (\gamma-1)\delta\left\{ma^{(\gamma-1)/\gamma} + (1-m)b^{(\gamma-1)/\gamma}\right\}\left\{mc^{(\gamma-1)/\gamma} + (1-m)e^{(\gamma-1)/\gamma}\right\}. \end{aligned} \tag{17}$$

Proof. Set $f(x) = x^{(\gamma-1)/\gamma}$. The function is strictly concave on $\mathbb{R}_+$, so for $0 < x < z < y$ its secant slopes satisfy

$$\frac{f(z)-f(x)}{z-x} > \frac{f(y)-f(x)}{y-x} > \frac{f(y)-f(z)}{y-z}. \tag{18}$$

The stationarity relations in the lemma imply

$$ae - bc = \delta, \qquad ac = bc + c\delta, \qquad be = bc + b\delta.$$

Thus ac and be both lie strictly between bc and ae. Apply (18), together with its endpoint limits $f'(bc) > [f(ae)-f(bc)]/(ae-bc) > f'(ae)$, once with $z = ac$ and once with $z = be$. Multiplying the resulting inequalities by m^2 and $(1-m)^2$, respectively, using $xf'(x) = [(\gamma-1)/\gamma]f(x)$, and collecting the two endpoint terms gives

$$\begin{aligned} &\frac{\gamma-1}{\gamma}\delta\left\{mf(a) + (1-m)f(b)\right\}\left\{mf(c) + (1-m)f(e)\right\} \\ &\quad > m(1-m)\left\{1 - \frac{\delta^2}{\gamma}\right\}\left\{f(ae) - f(bc)\right\}. \end{aligned}$$

Here the product on the first line follows from the exact expansion

$$\begin{aligned} &\left\{mf(a) + (1-m)f(b)\right\}\left\{mf(c) + (1-m)f(e)\right\} \\ &\quad = m^2 f(ac) + (1-m)^2 f(be) + m(1-m)\{f(ae) + f(bc)\}. \end{aligned}$$

Multiplying the strict inequality by γ yields (17). □

Proof of Proposition 3. The denominator $1 - \mathcal{E}\mathcal{H}$ is strictly positive because $\mathcal{E} > 0$ and $\mathcal{H} < 0$. In the benchmark economy, both $\mathcal{F}^{direct}$ and $\mathcal{F}^{indirect}$ equal zero, so $d\phi_{A,t+1}/d\phi_{A,t} = 0$. In the IGP-$\tau$ economy, stationarity and the definition of $\mathcal{F}^{direct}$ imply

$$\begin{aligned}
\phi_{A,t+1}(A) &= \phi_{A,t+1} + \phi_{N,t+1}\mathcal{F}^{direct}, \\
\phi_{A,t+1}(N) &= \phi_{A,t+1}(1 - \mathcal{F}^{direct}), \\
\phi_{N,t+1}(A) &= \phi_{N,t+1}(1 - \mathcal{F}^{direct}), \\
\phi_{N,t+1}(N) &= \phi_{N,t+1} + \phi_{A,t+1}\mathcal{F}^{direct}.
\end{aligned}$$

With inherited immobility, equation (2) gives $h_{j,t+1}(k) = \bar{\gamma}\phi_{j,t+1}(k)^{-1/\gamma}$. Hence effective labor supply is

$$\begin{aligned}
H_{A,t+1} &= \bar{\gamma}\left\{\phi_{A,t+1}\phi_{A,t+1}(A)^{(\gamma-1)/\gamma} + \phi_{N,t+1}\phi_{A,t+1}(N)^{(\gamma-1)/\gamma}\right\}, \\
H_{N,t+1} &= \bar{\gamma}\left\{\phi_{A,t+1}\phi_{N,t+1}(A)^{(\gamma-1)/\gamma} + \phi_{N,t+1}\phi_{N,t+1}(N)^{(\gamma-1)/\gamma}\right\}.
\end{aligned}$$

Under homothetic demand, substitution into the definitions of $\mathcal{F}^{indirect}$ and $\mathcal{H}$, followed by collection over the common denominator $H_{A,t+1}H_{N,t+1}$, gives

$$\begin{aligned}
&(1 - \mathcal{E}\mathcal{H})\,\mathcal{F}^{direct} - \phi_{A,t+1}\phi_{N,t+1}\mathcal{E}\mathcal{F}^{indirect} \\
&= \mathcal{F}^{direct}\left(1 + \frac{\gamma - 1}{\sigma}\right) \\
&\quad - \frac{\bar{\gamma}^2\phi_{A,t+1}\phi_{N,t+1}}{\sigma H_{A,t+1}H_{N,t+1}}\left\{\gamma - (\mathcal{F}^{direct})^2\right\} \\
&\quad \times \left\{[\phi_{A,t+1}(A)\phi_{N,t+1}(N)]^{(\gamma-1)/\gamma} - [\phi_{A,t+1}(N)\phi_{N,t+1}(A)]^{(\gamma-1)/\gamma}\right\}.
\end{aligned} \tag{19}$$

Apply Lemma 4 using the four stationary relations above. It gives the strict bound

$$\begin{aligned}
&\frac{\bar{\gamma}^2\phi_{A,t+1}\phi_{N,t+1}}{H_{A,t+1}H_{N,t+1}}\left\{\gamma - (\mathcal{F}^{direct})^2\right\} \\
&\quad \times \left\{[\phi_{A,t+1}(A)\phi_{N,t+1}(N)]^{(\gamma-1)/\gamma} - [\phi_{A,t+1}(N)\phi_{N,t+1}(A)]^{(\gamma-1)/\gamma}\right\} < (\gamma - 1)\mathcal{F}^{direct}.
\end{aligned}$$

Equation (19) is consequently strictly greater than $\mathcal{F}^{direct} > 0$. Since

$$\frac{d\phi_{A,t+1}}{d\phi_{A,t}} = \frac{(1 - \mathcal{E}\mathcal{H})\mathcal{F}^{direct} - \phi_{A,t+1}\phi_{N,t+1}\mathcal{E}\mathcal{F}^{indirect}}{1 - \mathcal{E}\mathcal{H}},$$

the denominator is positive and the IGP-τ derivative is strictly positive.

Finally, the total-response expression implies that the IGP-κ response is negative if and only if

$$\phi_{A,t+1}\phi_{N,t+1}\mathcal{E}^{\text{IGP-}\kappa}\mathcal{F}^{indirect,\text{IGP-}\kappa} > \left(1-\mathcal{E}^{\text{IGP-}\kappa}\mathcal{H}^{\text{IGP-}\kappa}\right)\mathcal{F}^{direct,\text{IGP-}\kappa}.$$

In this case, the induced relative-wage effect dominates the direct composition effect. If the inequality is reversed, the direct composition effect dominates and the derivative is positive; equality gives a zero response. Hence the model does not impose an unconditional sign on the derivative in the IGP-κ economy. □

C. Appendix: Quantitative Application

C.1. Quantitative Implementation and Additional Results

This appendix documents the data, production-side calibration, model inversion, counterfactual construction, and channel decomposition used in the quantitative analysis.

Data and timing. The quantitative exercise covers 1880, 1900, 1920, and 1940. The period-t cohort consists of ag-born and nonag-born workers with shares $\phi_{A,t-1}$ and $1-\phi_{A,t-1}$, respectively. Thus the agricultural employment share in 1860, $\phi_{A,0} = 0.665393$, determines cohort composition in 1880; thereafter, the observed aggregate share in one census determines the composition used for the next census. Table 15 describes the inputs and their sources. Aggregate and subgroup employment shares are constructed from the linked censuses as described in Section 1. Agricultural output shares, relative prices, and real GDP per capita are from Dennis and İşcan (2009), Alvarez-Cuadrado and Poschke (2011), and Maddison (1995), respectively. I normalize the latter two series to one in 1880.

Table 15: Data Used in the Quantitative Analysis

Variable	Description and use	Source or construction
$\phi_{A,0}$	Initial share of ag-born workers entering in 1880; determines initial cohort composition.	1860 agricultural employment share in the linked census sample.
$\phi_{A,t}$	Aggregate agricultural employment share; targeted by the inversion and used as the next period's ag-born share.	Linked census sample, 1880–1940.
$\phi_{A,t}(A)$, $\phi_{A,t}(N)$	Agricultural employment shares of ag-born and nonag-born workers; untargeted validation moments.	Linked census sample, 1880–1940.
$\omega_{A,t}$	Agricultural output share; calibrates and evaluates the production-side parameters.	Dennis and İşcan (2009).
$p_t = p_{N,t}/p_{A,t}$	Price of nonagricultural relative to agricultural goods; targeted by the inversion.	Alvarez-Cuadrado and Poschke (2011); indexed to one in 1880.
$Q_t = Y_t/P_t$	Real GDP per capita; targeted by the inversion.	Maddison (1995); indexed to one in 1880.

Note: Employment measures use the sample and sector definitions described in Section 1.

Production-side calibration. I set $\sigma = 0.3$, normalize $\Omega_N = 1$ and $\vartheta_A = 1$, and choose

$$\Omega_A = \frac{\omega_{A,1880}}{1 - \omega_{A,1880}} = 0.449275$$

so that the model exactly matches the 1880 agricultural output share. For any $\vartheta_N > 1$, the implied share is

$$\widehat{\omega}_{A,t}(\vartheta_N) = \frac{R_t(\vartheta_N)}{1 + R_t(\vartheta_N)}, \qquad R_t(\vartheta_N) = \Omega_A p_t^{\sigma-1} Q_t^{(1-\vartheta_N)(1-\sigma)},$$

where $p_t = p_{N,t}/p_{A,t}$ and $Q_t = Y_t/P_t$. I estimate

$$\widehat{\vartheta}_N = \arg\min_{\vartheta_N > 1} \sum_{t \in \{1900,1920,1940\}} \left[\omega_{A,t} - \widehat{\omega}_{A,t}(\vartheta_N)\right]^2,$$

obtaining $\widehat{\vartheta}_N = 4.630$. The optimization uses the transformation $\vartheta_N = 1 + \exp(u)$ and multiple starting values; the reported fit is the solution with the smallest sum of squared residuals.

Period-by-period model inversion. I now provide the complete inversion underlying Proposition 4.

Step 1: Worker allocations and effective labor. Combining equations (1) and (3) implies that the product of the relative payoff shifter and relative wage must satisfy the following equation.

$$\phi_{A,t} = \frac{\phi_{A,t-1}}{1 + \left[\frac{B_{N,t}}{B_{A,t}} \frac{w_{N,t}}{w_{A,t}} \frac{1-\tau}{\kappa_A}\right]^{\gamma}} + \frac{1-\phi_{A,t-1}}{1 + \left[\frac{B_{N,t}}{B_{A,t}} \frac{w_{N,t}}{w_{A,t}} \kappa_N\right]^{\gamma}}. \tag{20}$$

The right-hand side of equation (20) is continuous and strictly decreasing in the positive product $(B_{N,t}/B_{A,t})(w_{N,t}/w_{A,t})$, with limits one as the product approaches zero and zero as the product approaches infinity. Therefore, every observed $\phi_{A,t} \in (0,1)$ uniquely identifies this product and the two subgroup employment shares $\phi_{A,t}(A)$ and $\phi_{A,t}(N)$. Equations (2) and (4) then identify the following effective labor supplies.

$$H_{A,t} = \bar{\gamma}\left[\phi_{A,t-1}\kappa_A \phi_{A,t}(A)^{(\gamma-1)/\gamma} + (1-\phi_{A,t-1})\phi_{A,t}(N)^{(\gamma-1)/\gamma}\right], \tag{21}$$

$$H_{N,t} = \bar{\gamma}\left[\phi_{A,t-1}(1-\phi_{A,t}(A))^{(\gamma-1)/\gamma} + (1-\phi_{A,t-1})\kappa_N(1-\phi_{A,t}(N))^{(\gamma-1)/\gamma}\right]. \tag{22}$$

Both effective labor supplies are finite and strictly positive because $\gamma > 1$ and the subgroup employment shares are interior.

Step 2: Relative technology. Market clearing and sectoral production imply $X_{j,t} = Z_{j,t}H_{j,t}$. Substituting sectoral production into equation (10) gives the following condition.

$$\frac{Z_{A,t}H_{A,t}}{Z_{N,t}H_{N,t}} = \frac{\Omega_A}{\Omega_N} p_t^{\sigma} Q_t^{(\vartheta_A - \vartheta_N)(1-\sigma)}. \tag{23}$$

Rearranging this condition uniquely identifies the relative technology.

$$\frac{Z_{N,t}}{Z_{A,t}} = \frac{H_{A,t}}{H_{N,t}} \frac{\Omega_N}{\Omega_A} p_t^{-\sigma} Q_t^{(\vartheta_N - \vartheta_A)(1-\sigma)}. \tag{24}$$

Step 3: Technology levels. Using market clearing, the identified relative technology, and the observed Q_t in equation (8), I recover the sectoral technologies as follows.

$$Z_{A,t} = \left[\Omega_A^{1/\sigma}\left(\frac{H_{A,t}}{Q_t^{\vartheta_A}}\right)^{(\sigma-1)/\sigma} + \Omega_N^{1/\sigma}\left(\frac{(Z_{N,t}/Z_{A,t})H_{N,t}}{Q_t^{\vartheta_N}}\right)^{(\sigma-1)/\sigma}\right]^{-\sigma/(\sigma-1)}, \tag{25}$$

$$Z_{N,t} = \frac{Z_{N,t}}{Z_{A,t}} Z_{A,t}. \tag{26}$$

The expression in brackets is finite and strictly positive, so both recovered technologies are finite, strictly positive, and unique.

Step 4: Relative payoff shifter. Under the agricultural-good numeraire, equation (7) determines the relative wage.

$$\frac{w_{N,t}}{w_{A,t}} = p_t \frac{Z_{N,t}}{Z_{A,t}}.$$

Combining the identified relative wage with the nonag-born employment share in equation (1) identifies the relative payoff shifter.

$$\frac{B_{N,t}}{B_{A,t}} = \left(\frac{1-\phi_{A,t}(N)}{\phi_{A,t}(N)}\right)^{1/\gamma} \frac{1}{\kappa_N p_t (Z_{N,t}/Z_{A,t})}. \tag{27}$$

Only the relative payoff shifter is identified because multiplying $B_{A,t}$ and $B_{N,t}$ by a common positive constant does not change any worker's sector choice.

Proof of Proposition 4. Equation (20) uniquely identifies the product of the relative payoff shifter and relative wage, which identifies the subgroup employment shares and the positive effective labor supplies in equations (21) and (22). Equation (24) then uniquely identifies the relative technology. Given the relative technology, equations (25) and (26) uniquely identify both technology levels. Finally, equation (27) uniquely identifies the relative payoff shifter. Each mapping is single-valued and maps the stated interior observables into strictly positive fundamentals, establishing the result. □

Recursive implementation and model fit. Starting from the observed or imposed initial condition $\phi_{A,0}$, I apply the inversion sequentially because the recovered allocation in period t supplies the inherited composition $\phi_{A,t}$ used in period $t+1$. In the historical inversion, I use the observed $\phi_{A,t}$, rather than its fitted value, as the next period's inherited composition. At every date, the numerical implementation reconstructs the three

targeted outcomes and the CES aggregator with absolute errors below 10^{-8}. The agricultural output share is not an additional identifying moment in this inversion. Once p_t and Q_t are imposed, equation (11) determines the agricultural output share from the calibrated production-side parameters. The discrepancy from the observed output share therefore remains the production-side calibration residual rather than a fourth equation in the period-by-period inversion.

Untargeted subgroup employment shares. The inversion also implies $\phi_{A,t}(A)$ and $\phi_{A,t}(N)$, but does not target their observed counterparts. Conditional on $\phi_{A,t-1}$ and the time-invariant labor-supply parameters, the single aggregate target $\phi_{A,t}$ identifies the common payoff-wage term in equation (20), which in turn fixes both subgroup shares. Their comparison with the data in Figure 10 is therefore an overidentifying check. The labor-supply wedges are estimated from 1940 income and sector-choice data, so the model does not reproduce the substantially greater subgroup persistence observed in 1880. Matching these subgroup paths would require time-varying wedges or an additional model mechanism.

Counterfactual IGP regimes. Let superscript b denote the benchmark inversion and define the benchmark growth factors

$$g_t^A = \frac{Z_{A,t}^b}{Z_{A,1880}^b}, \qquad g_t^N = \frac{Z_{N,t}^b}{Z_{N,1880}^b}, \qquad g_t^B = \frac{(B_N/B_A)_t^b}{(B_N/B_A)_{1880}^b}.$$

I consider the benchmark regime, no inherited productivity ($\kappa_A = \kappa_N = 1$), no inherited immobility ($\tau = 0$), and no IGP ($\kappa_A = \kappa_N = 1, \tau = 0$). For each regime r, I first apply the inversion under that regime's parameters to the common 1880 targets. This produces regime-specific anchors $Z_{A,1880}^r$, $Z_{N,1880}^r$, and $(B_N/B_A)_{1880}^r$. The subsequent fundamentals are

$$Z_{A,t}^r = Z_{A,1880}^r g_t^A, \qquad Z_{N,t}^r = Z_{N,1880}^r g_t^N, \qquad (B_N/B_A)_t^r = (B_N/B_A)_{1880}^r g_t^B.$$

I then solve the equilibrium forward, using the simulated $\phi_{A,t}^r$ as the parental composition of the next cohort. This construction makes all regimes match the 1880 agricultural employment share, relative price, and real GDP per capita, while exposing them to the same proportional growth in fundamentals thereafter. Worker reallocation is $100(\phi_{A,1880}^r - \phi_{A,1940}^r)$ percentage points.

Inferred Drivers Without Inherited Productivity or Inherited Immobility Figure 15 separates the effect of omitting inherited productivity from the effect of omitting inherited immobility in the baseline inversion. The comparison uses the baseline values of $(\kappa_A, \kappa_N, \tau)$ whenever a wedge remains active and changes no other parameter.

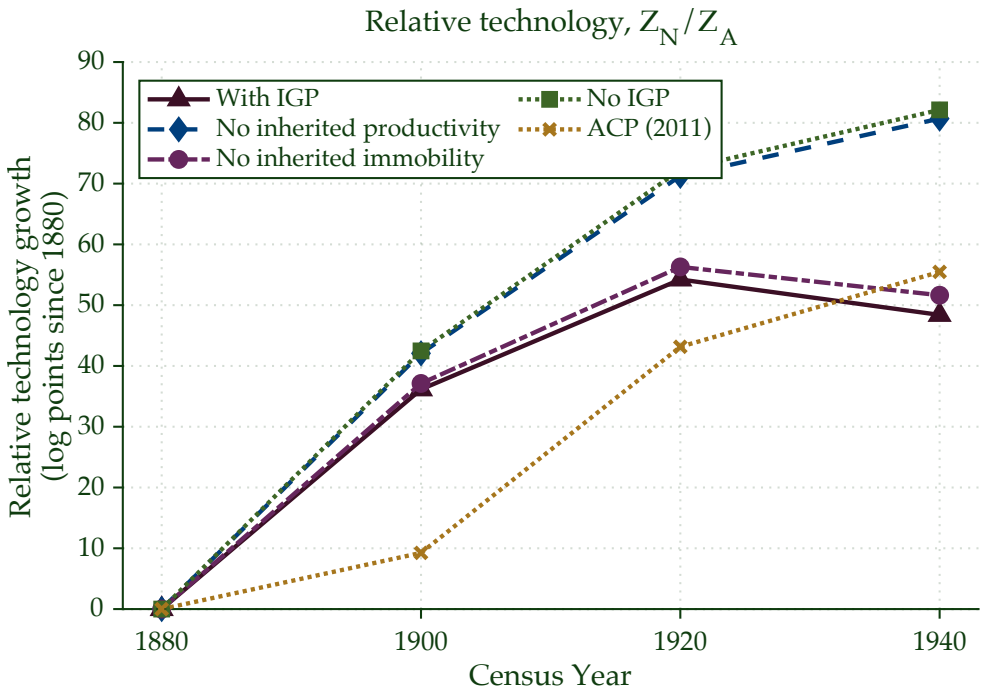


Figure 15: Inferred Relative Technology under Alternative IGP Restrictions

Note: The figure plots the cumulative log change in $Z_{N,t}/Z_{A,t}$ under the baseline IGP wedges, after setting only $\kappa_A = \kappa_N = 1$, after setting only $\tau = 0$, and after imposing both restrictions. The remaining wedges, γ, and all production-side parameters stay at their baseline values in each inversion. Every inversion matches the same agricultural employment share, relative price, and real GDP per capita in each year. The Alvarez-Cuadrado–Poschke series is the same nonfarm-to-farm relative TFP benchmark shown in Figure 12.

Under the baseline calibration, removing inherited productivity raises 1880–1940 relative-technology growth from 48.4 to 80.7 log points, whereas removing inherited immobility alone raises it to 51.6 log points and removing both wedges raises it to 82.1 log points. Thus the faster relative-technology growth inferred when IGP is omitted is driven primarily by the inherited-productivity wedges.

Interval-level Shapley decomposition. For regime r and an interval ending in t, let F_0^r and F_1^r denote the beginning- and end-of-interval fundamentals, and let M_0^r and M_1^r denote the corresponding inherited parental compositions. Write

$$y_{ab}^r = \phi_A(F_a^r, M_b^r), \qquad a, b \in \{0, 1\}.$$

The two-player Shapley contributions to the decline in the agricultural employment share are

$$C_t^r = \frac{1}{2}\left[(y_{00}^r - y_{10}^r) + (y_{01}^r - y_{11}^r)\right],$$
$$P_t^r = \frac{1}{2}\left[(y_{00}^r - y_{01}^r) + (y_{10}^r - y_{11}^r)\right],$$

where C_t^r is the contemporaneous response to technologies and the relative payoff shifter and P_t^r is intergenerational propagation through parental composition. The averages span the two possible orders in which the inputs change. They satisfy

$$C_t^r + P_t^r = y_{00}^r - y_{11}^r$$

in every interval. I sum each channel over 1880–1900, 1900–1920, and 1920–1940, then report its counterfactual-minus-benchmark difference in percentage points:

$$100 \sum_t (C_t^r - C_t^b), \qquad 100 \sum_t (P_t^r - P_t^b).$$

Their sum equals the difference between counterfactual and benchmark worker reallocation reported in Table 5.

C.1.1. Robustness to Alternative Calibrations

I evaluate the robustness of the quantitative results to alternative calibrations of the production-side and labor-supply parameters. Each robustness exercise repeats the entire procedure: re-invert the historical fundamentals under the alternative parameters, re-anchor every IGP regime to the common 1880 targets, simulate forward using that specification's inferred growth factors, and recompute the interval-level Shapley decomposition. The CLM exercise changes only the production-side parameters to $(\sigma, \vartheta_A, \vartheta_N) = (0.33, 1, 1/0.33)$. The added-fixed-effect labor-supply exercise retains the baseline production parameters and uses

$$(\gamma, \kappa_A, \kappa_N, \tau) = (2.288, 1.323, 1.437, 0.281).$$

The topcode-adjusted exercise likewise retains the baseline production parameters and uses

$$(\gamma, \kappa_A, \kappa_N, \tau) = (2.063, 1.615, 1.801, 0.328).$$

Thus the robustness results incorporate both the direct effect of alternative parameters and the associated changes in the inferred historical fundamentals; they do not hold the benchmark fundamental paths fixed in levels or growth rates.

Table 16: Robustness of Transmission Channels under Counterfactual IGP Regimes

IGP regime	Contemporaneous response	Intergenerational propagation	Total
	Percentage points		
Panel A: CLM production-side calibration			
No inherited productivity	2.71	3.39	6.10
No inherited immobility	3.28	-2.48	0.79
No intergenerational persistence	3.82	1.80	5.61
Panel B: Added-FE labor-supply calibration			
No inherited productivity	1.19	2.39	3.59
No inherited immobility	2.15	-1.67	0.49
No intergenerational persistence	2.00	1.19	3.19
Panel C: Topcode-adjusted labor-supply calibration			
No inherited productivity	2.64	3.36	5.99
No inherited immobility	3.24	-2.45	0.79
No intergenerational persistence	3.72	1.80	5.52

Note: Each panel repeats the interval-level Shapley decomposition under an alternative calibration. Panel A replaces the production parameters with the CLM mapping: $\sigma = 0.33$, $\vartheta_A = 1$, and $\vartheta_N = 1/0.33$. Panels B and C retain the baseline production parameters and use, respectively, the added-fixed-effect estimates $(\gamma, \kappa_A, \kappa_N, \tau) = (2.288, 1.323, 1.437, 0.281)$ and topcode-adjusted estimates $(2.063, 1.615, 1.801, 0.328)$. Each specification re-infers the historical fundamentals and re-anchors every counterfactual to the common 1880 targets; the two channels sum to the row total.

Table 16 shows that the channel results are qualitatively stable across the alternative calibrations. Removing inherited productivity raises reallocation through both channels, whereas removing inherited immobility produces a large contemporaneous response that is substantially offset by intergenerational propagation.

Table 17: Relative-Technology Growth under Alternative Calibrations and IGP Restrictions

IGP regime	Full-IGP growth	Counterfactual growth	Difference from full IGP
		1880–1940 log points	
Panel A: CLM production-side calibration			
No inherited productivity	−28.27	4.02	32.29
No inherited immobility	−28.27	−25.04	3.23
No IGP	−28.27	5.46	33.73
Panel B: Added-FE labor-supply calibration			
No inherited productivity	52.62	73.49	20.87
No inherited immobility	52.62	55.36	2.74
No IGP	52.62	74.68	22.06
Panel C: Topcode-adjusted labor-supply calibration			
No inherited productivity	48.69	81.22	32.53
No inherited immobility	48.69	51.90	3.21
No IGP	48.69	82.64	33.95

Note: Full-IGP and counterfactual growth are the cumulative 1880–1940 log changes in $Z_{N,t}/Z_{A,t}$ under the indicated calibration. "No inherited productivity" sets $\kappa_A = \kappa_N = 1$, "No inherited immobility" sets $\tau = 0$, and "No IGP" imposes both restrictions. The final column equals counterfactual growth minus full-IGP growth within the same panel. The three panels use the same alternative production-side and labor-supply calibrations as Table 16, and every row re-inverts the same observed employment, relative-price, and real-output paths.

Across these conventional calibration checks, removing all IGP raises inferred relative-technology growth by 22.1 to 33.9 log points relative to the corresponding full-IGP inversion.

I also assess whether the counterfactual results are sensitive to the skill distribution used to estimate the inherited-productivity and inherited-immobility wedges. For this exercise, I retain the baseline Fréchet model's γ and every baseline production-side parameter, but replace $(\kappa_A, \kappa_N, \tau)$ with the estimates from the bivariate lognormal and Fréchet–Frank specifications reported in Table 10. I then re-infer the historical fundamentals, re-anchor the counterfactual regimes to the common 1880 targets, and repeat the interval-level Shapley decomposition under each alternative wedge vector.

Table 18: Counterfactual Transmission Channels under Alternative Distribution-Implied Wedges

IGP regime	Contemporaneous response	Intergenerational propagation	Total
	Percentage points		
Panel A: Bivariate lognormal wedges			
No inherited productivity	0.98	1.18	2.16
No inherited immobility	3.77	−3.00	0.77
No intergenerational persistence	3.49	−1.49	2.00
Panel B: Fréchet–Frank wedges			
No inherited productivity	3.51	4.04	7.55
No inherited immobility	2.76	−2.28	0.48
No intergenerational persistence	4.23	2.90	7.13

Note: Each panel replaces only $(\kappa_A, \kappa_N, \tau)$ with the estimates implied by the indicated skill distribution and retains the baseline γ and production-side parameters. The historical fundamentals are re-inferred under each wedge vector, and every counterfactual is re-anchored to the common 1880 targets. The two channels sum to the row total, which is the change in 1880–1940 worker reallocation relative to the full-IGP benchmark under the same wedge vector.

Table 18 therefore isolates sensitivity to the way alternative skill distributions map the common targeted moments into inherited wedges rather than sensitivity to the rest of the macroeconomic model.

Table 19: Relative-Technology Growth under Alternative Distribution-Implied Wedges

IGP regime	Full-IGP growth	Counterfactual growth	Difference from full IGP
		1880–1940 log points	
Panel A: Bivariate lognormal wedges			
No inherited productivity	66.30	77.85	11.55
No inherited immobility	66.30	72.08	5.78
No IGP	66.30	82.13	15.83
Panel B: Fréchet–Frank wedges			
No inherited productivity	42.41	81.35	38.94
No inherited immobility	42.41	44.29	1.89
No IGP	42.41	82.13	39.72

Note: Full-IGP and counterfactual growth are the cumulative 1880–1940 log changes in $Z_{N,t}/Z_{A,t}$ under the indicated distribution-implied wedge estimates. "No inherited productivity" sets $\kappa_A = \kappa_N = 1$, "No inherited immobility" sets $\tau = 0$, and "No IGP" imposes both restrictions. The final column equals counterfactual growth minus full-IGP growth within the same panel. The panels replace only $(\kappa_A, \kappa_N, \tau)$ with the bivariate-lognormal or Fréchet–Frank estimates, retain the baseline γ and production-side parameters, and re-invert the same observed paths in every row.

The no-IGP increase is 15.8 log points under the bivariate-lognormal wedges and 39.7 log points under the Fréchet–Frank wedges, with inherited productivity again accounting for most of the difference in both cases.